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Basic Mathematics Used in Physics

1. Find the values of the following :-

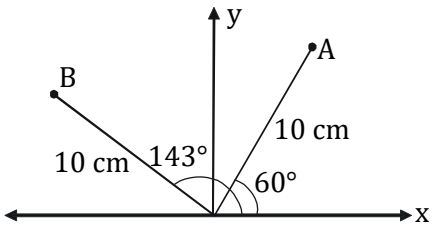
(a) sin 390° (b) cos 405°

(c) tan 420° (d) cos 450°

(e) $\sin\left(2\pi+\frac{\pi}{6}\right)$ (f) $\cos\left(2\pi+\frac{\pi}{3}\right)$

PBM181

2. Refer the given figure and identify correct statement(s)



(1) Distance of A from x-axis is $5\sqrt{3}$ cm.

(2) Distance of B from x-axis is 6 cm.

(3) Distance of A from y-axis is 5 cm.

(4) Distance of B from y-axis is 8 cm.

PBM182

3. The maximum and minimum values of expression (4–2 cos θ) respectively are :-

(1) 4 and 0 (2) 4 and 2

(3) 6 and 0 (4) 6 and 2

PBM183

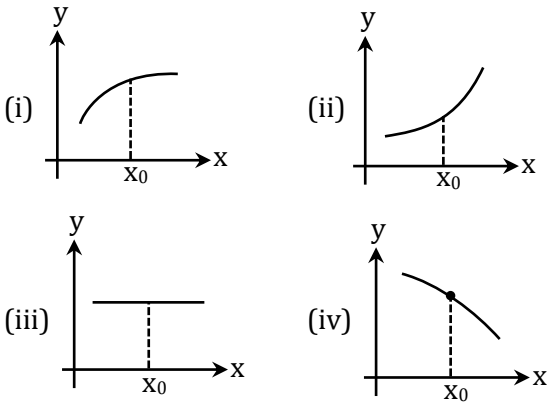
4. An airplane takes off at an angle 30° with the horizontal ground traveling at the speed of 180 km/h. If it continues to fly with the same velocity in the same direction, how long will it take to reach an altitude of 9 km above the ground?

(1) 5 minutes (2) 6 minutes

(3) 8 minutes (4) 9 minutes

PBM184

5. Which of the following statements are true based on graphs of y-versus x as shown below?



(1) Slope at x_0 is positive and non-zero in graph (i) and (ii)

(2) Slope is constant in graph (iii)

(3) Slope at x_0 is negative in graph (iv)

(4) Slope at x_0 is negative in graph (ii)

PBM185

6. Find the first derivatives

(i) If $y = x^{-1/3}$, find $\frac{dy}{dx}$

(ii) If $x = t^6$, find $\frac{dx}{dt}$

(iii) If $z = u^3$, find $\frac{dz}{du}$

(iv) If $f = r^2 + r^{1/3}$, find $\frac{df}{dr}$

(v) If $x = 2t^3 - 3t^2 + 5t - 6$, find $\frac{dx}{dt}$

(vi) If $y = 3x^2$

(vii) If $y = x \ln x$

(viii) If $y = x \sin x$

(ix) If $y = \sin x \cos x$

(x) If $y = \sin^2 x + \cos^2 x$

(xi) If $y = \cos 2x$

(xii) If $V = \frac{4}{3}\pi r^3$, find $\frac{dV}{dr}$

(xiii) If $S = 4\pi r^2$, find $\frac{dS}{dr}$

PBM186

7. If $y = \log_e x + \sin x + e^x$ then $\frac{dy}{dx}$ is :-

(1) $\frac{1}{x} + \sin x + e^x$ (2) $\frac{1}{x} - \cos x + e^x$

(3) $\frac{1}{x} + \cos x + e^x$ (4) $\frac{1}{x} - \sin x$

PBM187

8. $\frac{d}{dx}(e^{100}) =$ .....

(1) e^{100} (2) 0

(3) $_{100}e^{99}$ (4) None of these

PBM188

9. If $y = \sin x + \cos x$ then $\frac{d^2y}{dx^2}$ is :-

(1) sinx – cosx (2) cosx – sinx

(3) –(sinx + cosx) (4) None of these

PBM189

10. The radius of spherical bubble is changing with time. Then rate of change of its volume is given by :

(1) $4\pi r^2\frac{dr}{dt}$ (2) $\frac{4}{3}\pi r^2$

(3) $\frac{8}{3}\pi r^2$ (4) $\frac{8}{3}\pi r\frac{dr}{dt}$

PBM190

11. Given that $y=\frac{10}{\sin x+\sqrt{3}\cos x}$. Minimum value of y is :-

(1) zero (2) 2

(3) 5 (4) $10/\left(1+\sqrt{3}\right)$

PBM191

12. If $t=\sqrt{x}+4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is:

(1) 4 (2) Zero (3) 8 (4) 16

PBM192

13. How many solid spherical balls of radius 2cm can be casted (made) by melting a solid sphere of radius 8cm: -

(1) 4 (2) 16 (3) 32 (4) 64

PBM193

14. Evaluate the following indefinite integrals.

(a) $\int\frac{dx}{\sqrt{x}}$ (b) $\int\frac{dx}{(2x+3)}$

(c) $\int\sin x\cos x dx$

PBM194

15. Evaluate the following definite integrals.

(a) $\int_0^3(x^2+1)dx$

(b) $\int_2^3(x^3-4x^2+5x-10)dx$

(c) $\int_0^{\pi/4}(\cos x-\sin x)dx$

PBM195

16. Integrate the following

(i) $\int x^{-3/2}dx$

(ii) $\int\sin 60^\circ dx$

(iii) $\int\frac{1}{10x}dx$

(iv) $\int\left(2x^3-x^2+1\right)dx$

PBM196

17. Integrate the following :

(i) $\int\left(t-\frac{1}{t}\right)^2dt$ (ii) $\int\sin(10t-50)dt$

(iii) $\int e^{(100t+6)}dt$

PBM197

18. An object initially at rest moves along x–axis subjected to an acceleration which varies with time according to relation a = 2t + 5. Its velocity after 2 seconds will be : –

(1) 18 ms^{-1} (2) 9 ms^{-1}

(3) 12 ms^{-1} (4) 14 ms^{-1}

PBM198

19. The value of integral $\int_2^4\frac{dx}{x}$ is :-

(1) 3 $\log_e 2$ (2) $\log_e 2$

(3) $\log_e 4$ (4) 2 $\log_e 8$

PBM199

20. Area bounded by curve y = sin x, with x-axis, when x varies from 0 to $\frac{\pi}{2}$ is :-

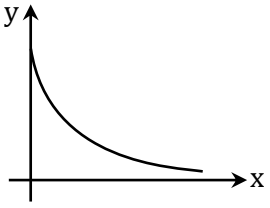
(1) 1 unit (2) 2 units

(3) 3 units (4) 0

PBM200

21. Which of the following equation is the best representation of the given graph's?

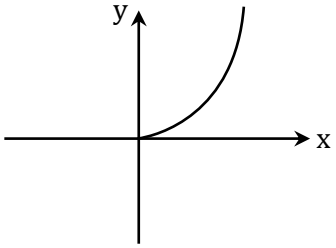
(i)



(1) $y=\frac{2}{x}$ (2) $y=e^{-x}$

(3) $y=\frac{1}{x^2}$ (4) $y=x^2$

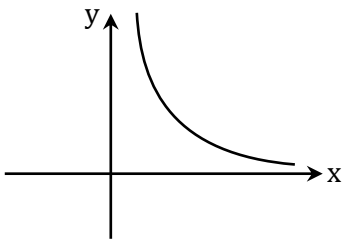
(ii)



(1) $y=x^2$ (2) $x=y^2$

(3) $y=e^x$ (4) $y=x$

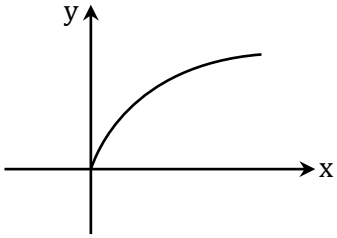
(iii)



(1) $y = e^{-x}$ (2) $y = e^{x}$

(3) $y = \frac{1}{x}$ (4) None of these

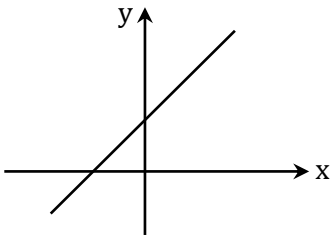
(iv)



(1) $y = 2x^2$ (2) $x = 2y^2$

(3) $y = -2x^2$ (4) $x = -2y^2$

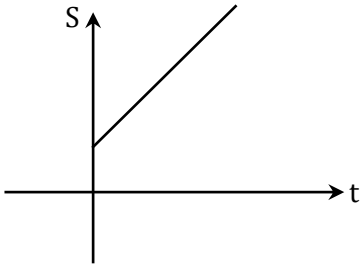
(v)



(1) $y = x + 1$ (2) $y = 2x - 1$

(3) $y = -x - 1$ (4) $y = -2x + 3$

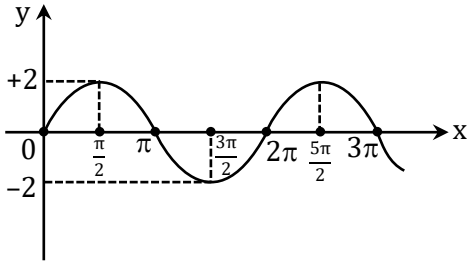
(vi)



(1) $S = 2t - 3$ (2) $S = 2t + 3$

(3) $S = 2t$ (4) None of these

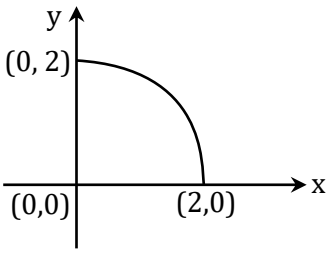
(vii)



(1) $y = 2\sin x$ (2) $y = 2\cos x$

(3) $y = \sin 2x$ (4) $y = \cos 2x$

(viii)



(1) $x + y = 2$ (2) $x^2 + y^2 = 4$

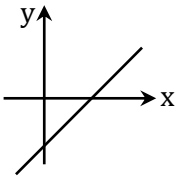
(3) $x^2 + y^2 = 2$ (4) $x^2 + y = 2$

PBM201

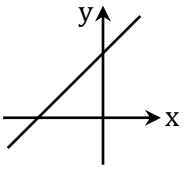
22. Which of two following graph is the best representation for the given equation.

(i) $y = 2x - 1$

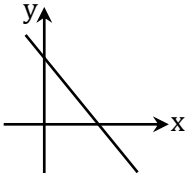
(1)



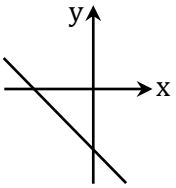
(2)



(3)

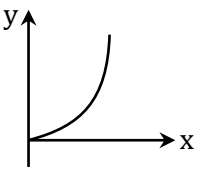


(4)

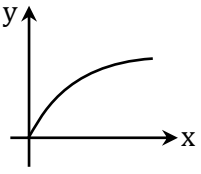


(ii) $y \propto x^2$

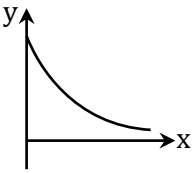
(1)



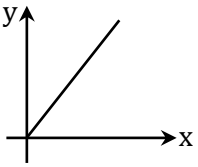
(2)



(3)

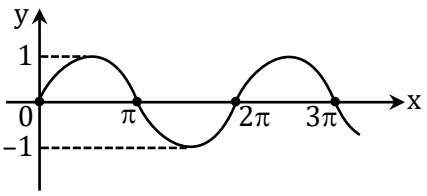


(4)

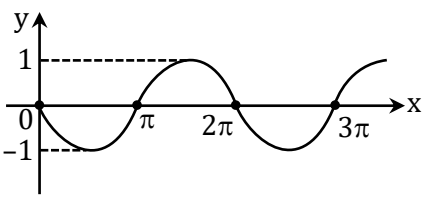


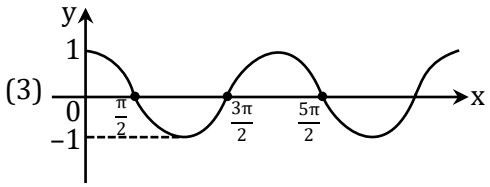
(iii) $y = \sin x$

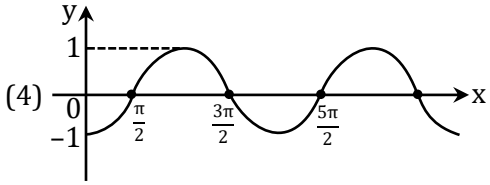
(1)



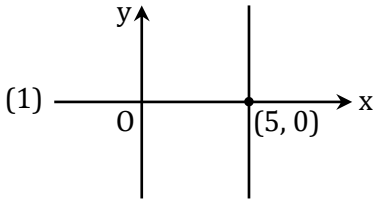
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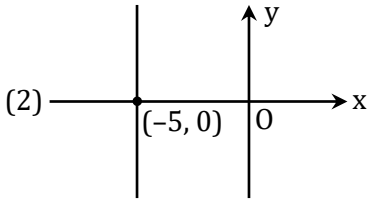


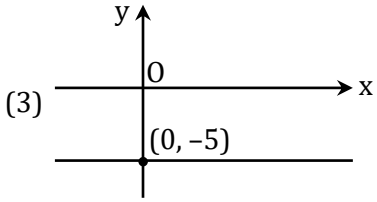
(3)


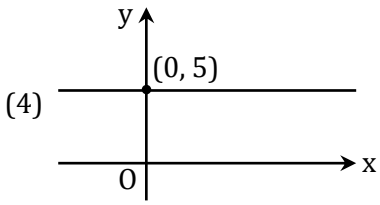
(4)


(iv) y = 5

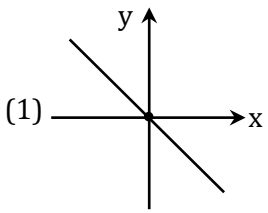
(1)


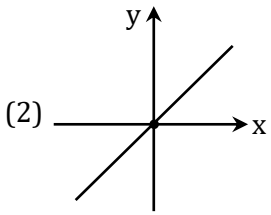
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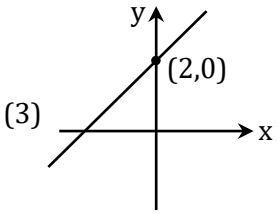
(3)


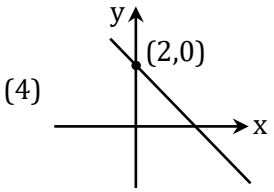
(4)


(v) y = 2x

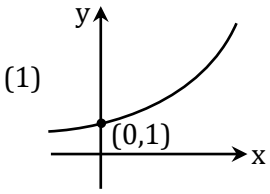
(1)


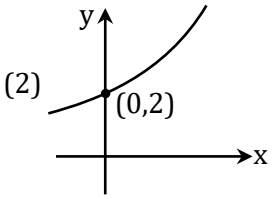
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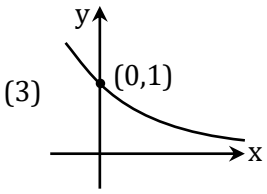
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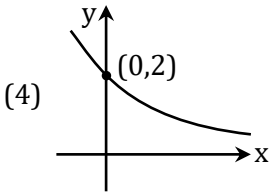
(4)


(vi) $y = 2e^x$

(1)


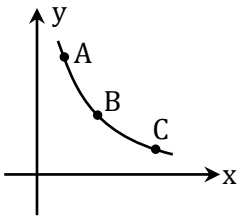
(2)


(3)


(4)


PBM202

23. The slope of graph in figure at point A, B and C is m_A, m_B and m_C respectively, then :-

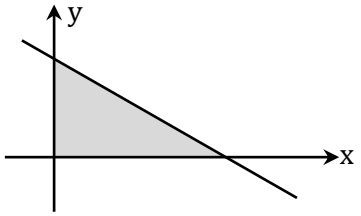


(1) $m_A > m_B > m_C$ (2) $m_A < m_B < m_C$

(3) $m_A = m_B = m_C$ (4) $m_A = m_C < m_B$

PBM203

24. In the given figure intercept on the positive y-axis made by straight line is 6 units and area enclosed by the line and both the axes is 72 square units. This area is shown by shaded portion in the figure. If both the axes have identical scale, equation of the line is :-



(1) $y=-\frac{1}{24}x+6$ (2) $y=-\frac{1}{2}x+6$

(3) $y=-\frac{1}{3}x+6$ (4) $y=-\frac{1}{4}x+6$

PBM204

25. Calculate the value of x in the following equation:-

(i) $x^2-5x+6=0$ (ii) $x^2-9x+20=0$

(iii) $x^2-15x+56=0$ (iv) $x^2-16x+63=0$

PBM205

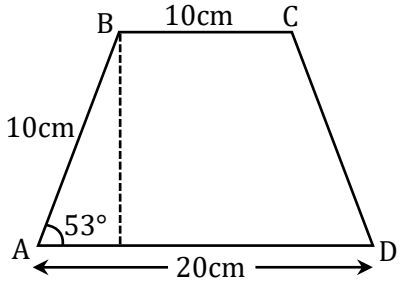
26. x-y relation is given by $y=x^n$. Calculate the value of n in the following :-

(i) $y=\frac{x^2}{x^{2/3}}$ (ii) $y=\frac{x^{3/2}}{x^1}$

(iii) $y=\frac{x^{1/2}}{x^{-1/2}}$ (iv) $y=\frac{x^{3/2}}{\sqrt{x}}$

PBM206

27. Find the area of shown trapezium :-



(1) 60 cm^2

(2) 120 cm^2

(3) 240 cm^2

(4) 360 cm^2

PBM207

ANSWER KEY

1. (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\sqrt{3}$ (d) 0 (e) $\frac{1}{2}$ (f) $\frac{1}{2}$

2. (1,2,3,4) **3.** (4) **4.** (2) **5.** (1,2,3)

6. (i) $-\frac{1}{3}x^{-4/3}$ (ii) $6t^5$ (iii) $3u^2$ (iv) $2r+\frac{1}{3}r^{-2/3}$ (v) $6t^2-6t+5$

(vi) 6x (vii) 1 + ln x (viii) x cos x + sin x (ix) $\cos^2x-\sin^2x$

(x) 0 (xi) $-2\sin 2x$ (xii) $4\pi r^2$ (xiii) $8\pi r$

7. (3) **8.** (2) **9.** (3) **10.** (1) **11.** (3) **12.** (2)

13. (4) **14.** (a) $2\sqrt{x}+C$ (b) $\frac{1}{2}\ell n(2x+3)+C$ (c) $-\frac{1}{4}\cos 2x+C$

15. (a) 12 (b) $-\frac{79}{12}$ (c) $\sqrt{2}-1$

16. (i) $-2x^{-1/2}+C$ (ii) $\frac{\sqrt{3}}{2}x+C$ (iii) $\frac{1}{10}\log x+C$ (iv) $\frac{x^4}{2}-\frac{x^3}{3}+x+C$

17. (i) $\frac{t^3}{3}-\frac{1}{t}-2t+C$ (ii) $-\frac{\cos(10t-50)}{10}+C$ (iii) $\frac{1}{100}e^{(100t+6)}+C$

18. (4) **19.** (2) **20.** (1)

21. (i) 2 (ii) 1 (iii) 3 (iv) 2 (v) 1 (vi) 2 (vii) 1 (viii) 2

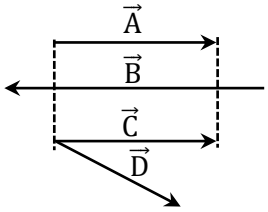
22. (i) 1 (ii) 1 (iii) 1 (iv) 4 (v) 2 (vi) 2

23. (2) **24.** (4) **25.** (i) 2,3, (ii) 4,5 (iii) 7,8 (iv) 7,9

26. (i) $\frac{4}{3}$ (ii) $\frac{1}{2}$ (iii) 1 (iv) 1 **27.** (2)

Vectors

1. State true or false for the given vectors



(i) $\vec{A}$ and $\vec{B}$ are opposite vectors.

(ii) $\vec{A}$ and $\vec{C}$ are equal as well as parallel.

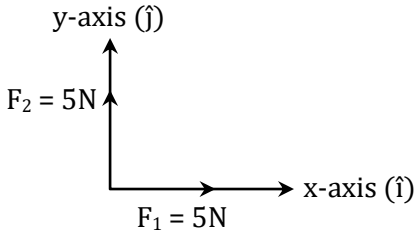
(iii) $\vec{A}$, $\vec{C}$ and $\vec{D}$ are coplanar vectors

(iv) $\vec{B}$, $\vec{C}$ and $\vec{D}$ are coplanar vectors

(v) $\vec{C}$ and $\vec{D}$ are coinitial vectors

PBM208

2. For the given two forces $\vec{F}_1$ & $\vec{F}_2$, state true or false



means $\vec{F}_1 = 5N\hat{i}$ $\vec{F}_2 = 5N\hat{j}$

(i) $\vec{F}_1 = \vec{F}_2$ (ii) $|\vec{F}_1| = |\vec{F}_2|$

(iii) $F_1 = F_2$ (iv) $\vec{F}_1 \perp \vec{F}_2$

(v) $\hat{F}_1 = \hat{F}_2$ (vi) $\hat{F}_1 \perp \hat{F}_2$

(vii) $|\hat{F}_1| = |\hat{F}_2|$

PBM209

3. State whether the following relations are true or false: -

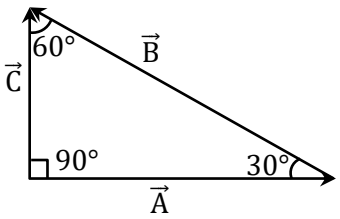
(i) $\overrightarrow{AB} = \overrightarrow{BA}$ (ii) $\overrightarrow{AB} = -\overrightarrow{BA}$

(iii) $|\overrightarrow{AB}| = |\overrightarrow{BA}|$ (iv) $|\overrightarrow{AB}| = |-\overrightarrow{AB}|$

(v) $\hat{j} = \hat{k}$ (vi) $|\hat{j}| = |\hat{k}|$

PBM210

4. For the given vector triangle find the following angle?



(i) Angle between $\vec{A}$ and $\vec{B}$ =

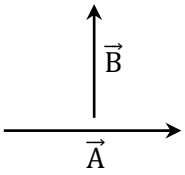
(ii) Angle between $\vec{B}$ and $\vec{C}$ =

(iii) Angle between $\vec{A}$ and $\vec{C}$ =

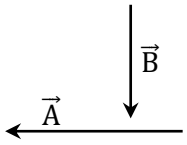
PBM211

5. Add vectors $\vec{A}$ & $\vec{B}$ by triangle law in the following cases :

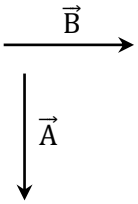
(i)



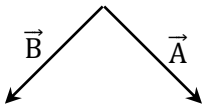
(ii)



(iii)

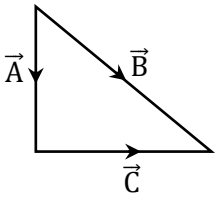


(iv)



PBM212

6. For the given vectors, which of the following option is correct :



(1) $\vec{A}+\vec{B}=\vec{C}$ (2) $\vec{A}+\vec{C}=\vec{B}$

(3) $\vec{B}+\vec{C}=\vec{A}$ (4) None of these

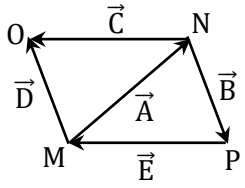
PBM213

7. Which of the following group of concurrent forces may be in equilibrium (R = 0)?

	A	B	C
(1)	10	20	40
(2)	3	5	1
(3)	20	20	20
(4)	40	30	5

PBM214

8. From the figure, the correct relation is -

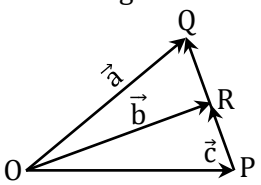


(1) $\vec{A}+\vec{B}+\vec{E}=\vec{0}$ (2) $\vec{C}-\vec{D}=-\vec{A}$

(3) $\vec{B}+\vec{E}-\vec{C}=-\vec{D}$ (4) All of the above

PBM215

9. Figure shows three vectors $\vec{a},\vec{b}$ and $\vec{c}$. If $\overrightarrow{RQ}=2\overrightarrow{PR}$, which of the following relation is correct: -



(1) $2\vec{a}+\vec{c}=3\vec{b}$

(2) $\vec{a}+3\vec{c}=2\vec{b}$

(3) $3\vec{a}+\vec{c}=2\vec{b}$

(4) $\vec{a}+2\vec{c}=3\vec{b}$

PBM216

10. The resultant of the vectors $\vec{A}$ and $\vec{B}$ is perpendicular to the vector $\vec{A}$ and its magnitude is equal to half the magnitude of vector $\vec{B}$. The angle between $\vec{A}$ and $\vec{B}$ is :-

(1) 120°

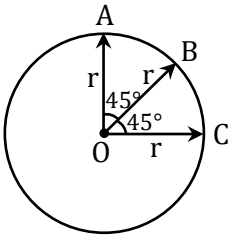
(2) 150°

(3) 135°

(4) None of these

PBM217

11. The resultant of the three vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ shown in figure :-



(1) r (2) 2r

(3) $r(1+\sqrt{2})$ (4) $r(\sqrt{2}-1)$

PBM218

12. If $\frac{|\vec{a}+\vec{b}|}{|\vec{a}-\vec{b}|}$ = 1, then angle between $\vec{a}$ and $\vec{b}$ is

(1) 0° (2) 45°

(3) 90° (4) 60°

PBM219

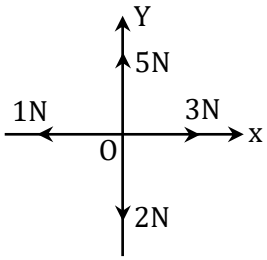
13. I started walking down a road to day-break facing the sun. After walking for some time, I turned to my left, then I turned to the right once again. In which direction was I going then :-

(1) East (2) North-west

(3) North-east (4) South

PBM220

14. In the shown figure what amount of force and in which direction we should apply so that net force only acts along x-axis only : -



(1) 3N along –ve of y-axis

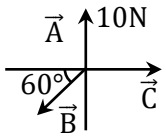
(2) 3N along +ve of y-axis

(3) 2N along –ve of x-axis

(4) 2N along +ve of x-axis

PBM221

15. If $\vec{A}+\vec{B}+\vec{C}=\vec{0}$ then $|\vec{B}|$ and $|\vec{C}|$ are :



(1) 10N, 20N (2) $\frac{20}{\sqrt{3}}$ N, 10N

(3) $\frac{20}{\sqrt{3}}$ N, $\frac{10}{\sqrt{3}}$ N (4) 20N, $\frac{10}{\sqrt{3}}$ N

PBM222

16. Resultant of two forces $\vec{F}_1$ and $\vec{F}_2$ has magnitude 50 N. The resultant is inclined to $\vec{F}_1$ at 60° and to $\vec{F}_2$ at 30°. Magnitudes of $\vec{F}_1$ and $\vec{F}_2$ respectively are: -

(1) 25 N, $25\sqrt{3}$ N

(2) 20 N, $20\sqrt{3}$ N

(3) 20 N, 30 N

(4) 30 N, 40 N

PBM223

17. The dot product of two vectors of magnitudes 3 units and 5 units cannot be :-

(i) – 20 (ii) 16 (iii) – 10 (iv) 14

(1) (i, iii) (2) (i, ii)

(3) (i, iv) (4) (ii, iii, iv)

PBM224

18. Which of the following relations is true for two unit vectors $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

(1) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\tan\frac{\theta}{2}$

(2) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\tan\frac{\theta}{2}$

(3) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\cos\frac{\theta}{2}$

(4) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\cos\frac{\theta}{2}$

PBM225

19. If $\left|\hat{A}\times\hat{B}\right|=-\sqrt{3}\,\hat{A}\cdot\hat{B}$, then $\left|\hat{A}-\hat{B}\right|=$.....

(1) 0 (2) 1

(3) –1 (4) $\sqrt{3}$

PBM226

20. If the vectors $\left(\hat{i}+\hat{j}+\hat{k}\right)$ and $3\hat{i}$ form two sides of a triangle, then area of the triangle is:

(1) $\sqrt{3}$ unit (2) $2\sqrt{3}$ unit

(3) $\frac{3}{\sqrt{2}}$ unit (4) $3\sqrt{2}$ unit

PBM227

21. Consider two vectors $\vec{F}_1=2\hat{i}+5\hat{k}$ and $\vec{F}_2=3\hat{j}+4\hat{k}$. The scalar product of these vectors is :-

(1) 20 (2) 23

(3) $5\sqrt{33}$ (4) 26

PBM228

22. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A}+\vec{B}$ is equal to two times the magnitude of $\vec{A}-\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

(1) $\sin^{-1}\left(\frac{3}{5}\right)$ (2) $\sin^{-1}\left(\frac{1}{3}\right)$

(3) $\cos^{-1}\left(\frac{3}{5}\right)$ (4) $\cos^{-1}\left(\frac{1}{3}\right)$

PBM229

23. If $\vec{A}=\left(2\hat{i}+3\hat{j}-\hat{k}\right)$ m and $\vec{B}=\left(\hat{i}+2\hat{j}+2\hat{k}\right)$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be_____ m.

PBM230

24. When $\vec{A}\cdot\vec{B}=-\left|\vec{A}\right|\left|\vec{B}\right|$, then :-

(1) $\vec{A}$ and $\vec{B}$ are perpendicular to each other

(2) $\vec{A}$ and $\vec{B}$ act in the same direction

(3) $\vec{A}$ and $\vec{B}$ act in the opposite direction

(4) $\vec{A}$ and $\vec{B}$ can act in any direction

PBM231

25. If $\vec{A}\times\vec{B}=\vec{C}$ then find out the correct one :-

(1) $\vec{A}.\vec{B}=0$ (2) $\vec{A}.\vec{C}\neq0$

(3) $\vec{B}.\vec{C}\neq0$ (4) $\left(\vec{A}+\vec{B}\right).\vec{C}=0$

PBM232

26. The position vectors of points A, B, C and D are $\vec{A}=3\hat{i}+4\hat{j}+5\hat{k}$, $\vec{B}=4\hat{i}+5\hat{j}+6\hat{k}$, $\vec{C}=7\hat{i}+9\hat{j}+3\hat{k}$ and $\vec{D}=4\hat{i}+6\hat{j}$, then displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are: -

(1) Same (2) Parallel

(3) Perpendicular (4) Antiparallel

PBM233

27. $\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non-zero constant. Which of the following expressions is true for $\vec{A}$?

(1) $\vec{A}.\vec{A}=0$ (2) $\vec{A}\times\vec{A}<0$

(3) $\vec{A}\times\vec{A}=0$ (4) $\vec{A}\times\vec{A}>0$

PBM234

28. If $\vec{P}=3\hat{i}+\sqrt{3}\hat{j}+2\hat{k}$ and $\vec{Q}=4\hat{i}+\sqrt{3}\hat{j}+2.5\hat{k}$ then, The unit vector in the direction of $\vec{P}\times\vec{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{i}+\hat{j}-2\sqrt{3}\hat{k}\right)$. The value of x is

PBM235

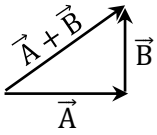
ANSWER KEY

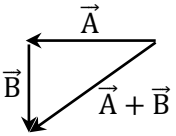
1. (i) False, (ii) True, (iii) True, (iv)True, (v) True

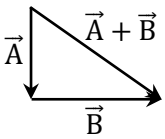
2. (i) False, (ii) True, (iii) True, (iv) True, (v) False, (vi) True, (vii) True

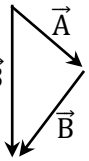
3. (i) False, (ii) True, (iii) True, (iv) True, (v) False, (vi) True

4. (i) 180° – 30°=150°, (ii) 60°, (iii) 90°

5. (i) 

, (ii) 

, (iii) 

, (iv) $\vec{A}+\vec{B}$ 

6.	(2)	**7.**	(3)	**8.**	(4)	**9.**	(4)	**10.**	(2)
11.	(3)	**12.**	(3)	**13.**	(1)	**14.**	(1)	**15.**	(3)
16.	(1)	**17.**	(2)	**18.**	(2)	**19.**	(4)	**20.**	(3)
21.	(1)	**22.**	(3)	**23.**	(2)	**24.**	(3)	**25.**	(4)
26.	(4)	**27.**	(3)	**28.**	(4)				

Kinematics

1. A man covered following displacements. Find the net displacement of the person
(i) 25 m west
(ii) 20 m 30° east of south
(iii) $5\sqrt{3}$ m south
(iv) 10 m east
(v) 10 m 60° north of west

PMP238

2. A person moves towards north with uniform speed of 15 m/s and cover 16 m. After this he moves towards east and covers 6 m and finally covers 8 m distance with same speed in south. Find out
(i) Distance travelled by person
(ii) Displacement of person
(iii) Average velocity of person

PMP239

3. A particle is moving with constant speed v on a circular path of 'r' radius when it has moved by angle 60°, Find
(i) Displacement of particle
(ii) Average velocity
(iii) Average acceleration

PMP240

4. A man travels half time of its journey with speed of 5 m/s and for remaining half time he moves half of its total distance travelled, with speed of 3 m/s and remaining half distance with speed of 6 m/s. Find out the average speed of the man?

PMP241

5. A car is moving with a velocity of 10 m/s. Suddenly driver sees a child at a distance of 30 m and he applies the brakes, it produced a uniform retardation in car and car stops in 5 second. Find (mass of car = 1000 kg)
(i) the stopping distance of car
(ii) will the car hit the child
(iii) the retardation in car
(iv) resistive force on car

PMP242

6. A particle starts from rest and moves with uniform acceleration. After n^{th} second its velocity is v then how much distance is travelled by it in last 2 second?

PMP243

7. A car starts from rest and moves with uniform acceleration for time t then it moves with uniform speed of 60 km/h for time 8t, finally it moves with uniform retardation for time 3t and comes to rest. Find the average speed of car in its total journey.

PMP244

8. Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t)=\alpha t+\beta t^2$ and $X_Q(t)=ft-t^2$. At what time, both the buses have same velocity ?

(1) $\frac{\alpha-f}{1+\beta}$

(2) $\frac{\alpha+f}{2(\beta-1)}$

(3) $\frac{\alpha+f}{2(1+\beta)}$

(4) $\frac{f-\alpha}{2(1+\beta)}$

PMP245

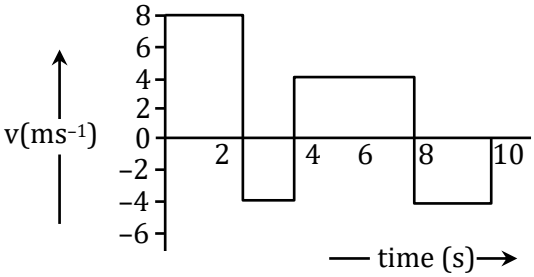
9. A particle is moving in a straight line such that its velocity is increasing at 5 ms^{-1} per meter. The acceleration of the particle is ____ ms^{-2} at a point where its velocity is 20 ms^{-1}.

PMP246

10. A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is :
(Given g = 9.8 m/s^2)
(1) 5 m (2) 10 m
(3) 19.8 m (4) 24.5 m

PMP247

11. The velocity time graph of a body moving in a straight line is shown in figure.



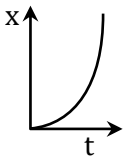
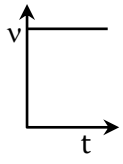
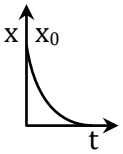
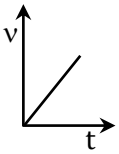
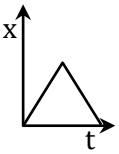
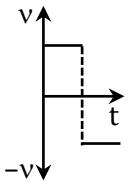
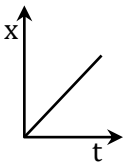
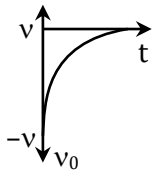
The ratio of displacement to distance travelled by the body in time 0 to 10s is

(1) 1:1 (2) 1:4

(3) 1:2 (4) 1:3

PMP248

12. Match Column-I with Column-II :

Column-I (x-t graphs)		Column-II (v-t graphs)	
A.		I.	
B.		II.	
C.		III.	
D.		IV.	

Choose the correct answer from the options given below:

(1) A- II B-IV, C-III, D-I

(2) A- I. B-II, C-III, D-IV

(3) A- II B-III, C-IV, D-I

(4) A- I, B-III. C-IV, D-II

PMP249

13. A body is thrown vertically upwards and takes 5 seconds to reach maximum height. The distance travelled by the body will be same in :-

(1) 1st and 10th second

(2) 2nd and 8th second

(3) 4st and 6th second

(4) Both (2) & (3)

PMP250

14. A ball is dropped from a height h above ground. Neglect the air resistance, its velocity (v) varies with its height (y) above the ground as :-

(1) $\sqrt{2g(h-y)}$ (2) $\sqrt{2gh}$

(3) $\sqrt{2gy}$ (4) $\sqrt{2g(h+y)}$

PMP251

15. A body projected vertically upwards, reaches a height of 180 m. Find out [g = 10 ms^{-2}]

(i) initial velocity of body

(ii) time taken by body to reach its maximum height

(iii) the time for which the body remains in air

(iv) height of the body from the ground after 8 sec.

(v) velocity of body at t = 8 sec

(vi) the time at which the body reach the height of 100 m

(vii) A man at height of 135 m from ground tries to catch the body but he misses. After how much time from now he will be able to catch the body again?

PMP252

16. A particle is projected vertically upwards from the earth. It crosses the same point at t = 2 sec and t = 8 sec. Find out [g = 10 ms^{-2}]

(i) the time for which particle remains in air

(ii) initial velocity of particle

(iii) maximum height attained by the particle

(iv) height of the particle at t = 2 sec and t = 8 sec from earth

PMP253

17. A body is dropped from the top of a tower and it covers a 80 m distance in last two seconds of its journey. Find out [g = 10 ms^{-2}]

(i) height of the tower

(ii) time taken by body to reach the ground

(iii) the velocity of body when it hits the ground

PMP254

18. Water drops are falling from a tap at regular time intervals. When the fifth drop is near to fall from tap the first drop is at ground. If the tap is fixed at height H from ground then find out

(i) the height of the second drop from ground

(ii) distance between the second and third drop

(iii) velocity of third drop

PMP255

19. From the top of a tower, a ball is thrown vertically upward which reaches the ground in 6 s. A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location, will reach the ground in ____ s.

PMP256

20. When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is :

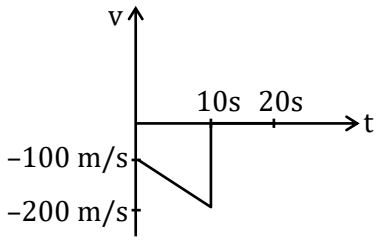
(1) 19.6 m (2) 29.4 m

(3) 39.2 m (4) 73.5 m

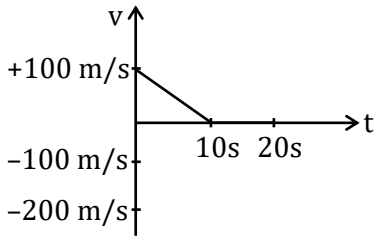
PMP257

21. A bullet is shot vertically downwards with an initial velocity of 100 m/s from a certain height. Within 10 s, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time t = 20 s will be : (Take g = 10 m/s^2)

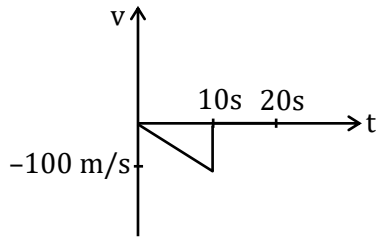
(1)



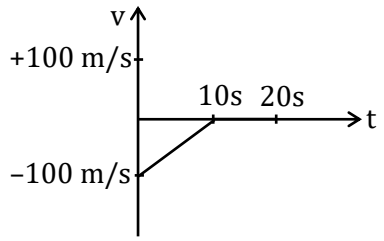
(2)



(3)



(4)



PMP258

22. A ball is thrown vertically upwards with a velocity of 19.6 ms^{-1} from the top of a tower. The ball strikes the ground after 6 s. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)$m. The value of k is ..... (use g = 9.8 m/s^2)

PMP259

23. A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

(1) g/2n (2) g/n

(3) 2gn (4) g/2n^2

PMP260

24. A ball is released from a height h. If t_1 and t_2 be the time required to complete first half and second half of the distance respectively. Then, choose the correct relation between t_1 and t_2.

(1) $t_1 = (\sqrt{2})t_2$ (2) $t_1 = (\sqrt{2}-1)t_2$

(3) $t_2 = (\sqrt{2}+1)t_1$ (4) $t_2 = (\sqrt{2}-1)t_1$

PMP261

25. A tennis ball is dropped on to the floor from a height of 9.8 m. It rebounds to a height 5.0 m. Ball comes in contact with the floor for 0.2s. The average acceleration during contact is ___ ms^{-2}. [Given g = 10 ms^{-2}]

PMP262

26. A ball is projected from ground in such a way that after 10 seconds of projection it lands on ground 500 m away from the point of projection. Find out :-
(i) angle of projection
(ii) velocity of projection
(iii) Velocity of ball after 5 seconds

PMP263

27. A body is projected at an angle of 45° with horizontal with velocity of $40\sqrt{2}$ m/s. Find out
(i) Maximum height attained by body
(ii) Time of flight
(iii) Horizontal range
(iv) Velocity at maximum height
(v) The ratio of Kinetic energy to potential energy at highest point
(vi) The path equation of projectile, assuming point of projection as origin and consider x and y are horizontal and vertical distance in meter
(vii) The horizontal distance covered by body in 2 seconds
(viii) The vertical distance covered by body in 2 second

PMP264

28. A ball is thrown from a corner of room of length 20 m and height 5 m such that the ball moves along the length of room just touching the ceiling and drops on the other corner. Find out
(i) Angle of projection
(ii) Velocity of projection
(iii) Time of flight

PMP265

29. A hunter aims his gun and fires a bullet directly towards a monkey sitting on a distance tree. At the instant the bullet leaves the barrel of the gun, the monkey drops from the tree :
(i) Will the bullet hit the aim?
(ii) What Path of bullet will be appeared to monkey?
(iii) If monkey does not drop from the tree will bullet hit the aim?

PMP266

30. A particle of mass 50 g is projected horizontally from the top of a tower of height 50 m with a velocity 20 m/s. If g = 10 m/s^2, then find the: -
(take downward as positive y direction)
(1) Velocity at the instant t = 2 s
(2) Position at the instant t = 3 s
(3) Velocity just before hitting the ground
(4) Change in velocity in 2 s

PMP267

31. A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1}. The maximum height reached by the body during its motion is _____m. (use g = 10ms^{-2})

PMP268

32. Two projectile thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is

(1) $1:\sqrt{2}$ (2) 2 : 1

(3) $\sqrt{2}:1$ (4) 1 : 2

PMP269

33. If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y=5x(1-x)$. The y component vector of the initial velocity is ____ $\hat{j}$ (Take $g=10 m/s^2$)

PMP270

34. A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point (20, 10). If T is the time of flight, then its momentum vector, at time $t=\frac{T}{\sqrt{2}}$, is _____

[Take g = 10 m/s^2]

(1) $100\hat{i}+(100\sqrt{2}-200)\hat{j}$

(2) $100\sqrt{2}\,\hat{i}+(100-200\sqrt{2})\hat{j}$

(3) $100\,\hat{i}+(100-200\sqrt{2})\hat{j}$

(4) $100\sqrt{2}\,\hat{i}+(100\sqrt{2}-200)\hat{j}$

PMP271

35. At time t = 0 a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping z coordinate constant. At any instant of time it's position along the x and y directions are defined as 3t and $5t^3$ respectively. At t = 1s acceleration of the particle will be

(1) $-30y$ (2) $30y$

(3) $3x+15y$ (4) $3x+15y+7\hat{z}$

PMP272

36. An object is projected in the air with initial velocity u at an angle θ. The projectile motion is such that the horizontal range R, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be _____degree.

PMP273

37. The initial speed of a projectile fired from ground is u. At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of the projectile is:

(1) $\frac{u}{2g}$ (2) $\frac{u}{g}$

(3) $\frac{2u}{g}$ (4) $\frac{\sqrt{3}u}{g}$

PMP274

38. L_1 metre long train is moving with a velocity of v_1 m/s and another L_2 metre long train is moving with a velocity of v_2 m/s in opposite direction. Find out (i) The time taken by two train to cross each other (ii) Distance travel by the each train during the crossing.

PMP275

39. A car is moving with a speed of 25 km/h in East and car driver observes that a bus is moving with a speed of $25\sqrt{3}$ km/h in the north what will be actual velocity of bus.

PMP276

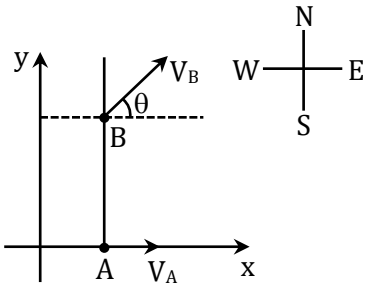
40. The wind is blowing in south with velocity of 4 km/h and observer is travelling with velocity of $4\sqrt{3}$ km/h in east. Find out the velocity of wind appeared to observer.

PMP277

41. Rain is falling vertically downwards with a velocity of 2 m/s. A person is running to the North with a velocity of $2\sqrt{3}$ m/s. Find out the velocity and direction of rain as appeared to the person.

PMP278

42. Figure shows two ships moving in x-y plane with velocities V_A and V_B. The ships move such that B always remains north of A. The ratio $\frac{V_A}{V_B}$ is equal to: -



(1) cos θ (2) sin θ

(3) sec θ (4) cosec θ

PMP279

43. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}\,kmh^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is :

(1) $30\,kmh^{-1}$ (2) $\frac{25}{\sqrt{2}}kmh^{-1}$

(3) $\frac{30}{\sqrt{2}}kmh^{-1}$ (4) $25kmh^{-1}$

PMP280

44. The speed of a swimmer is 4 km h^{-1} in still water. If the swimmer makes his strokes normal to the flow of river of width 1 km, he reaches a point 750 m down the stream on the opposite bank.

The speed of the river water is _____km h^{-1}.

PMP281

ANSWER KEY

1. 20 m, 60° south of west **2.** (i) 30 m, 53°, N of E, (ii) 10 m, (iii) 5 m/s 53°, N of E

3. (i) r, (ii) $\frac{3v}{\pi}$, (iii) $\frac{3v^2}{\pi r}$ **4.** 4.5 m/s

5. (i) 25 m, (ii) No, (iii) 2m/s^2, (iv) 2000 N **6.** $\frac{2v(n-1)}{n}$

7. 50 km/hr **8.** (4) **9.** 100 **10.** (1) **11.** (4)

12. (1) **13.** (1) **14.** (1)

15. (i) 60 m/s, (ii) 6 sec, (iii) 12 sec, (iv) 160 m, (v) –20 m/s, (vi) 2 sec, 10 sec, (vii) 6 sec

16. (i) 10 sec, (ii) 50 m/s, (iii) 125 m, (iv) 80 m **17.** (i) 125 m, (ii) 5 sec, (iii) 50 m/s

18. (i) $\frac{7}{16}$ H, (ii) $\frac{5}{16}$ H, (iii) $\sqrt{\frac{gH}{2}}$ **19.** (3) **20.** (2) **21.** (1)

22. 392 **23.** (4) **24.** (4) **25.** 120

26. (i) $\theta = 45°$, (ii) $50\sqrt{2}$ m/s, (iii) 50 m/sec

27. (i) 80 m, (ii) 8 sec, (iii) 320 m, (iv) 40 m/s, (v) 1 : 1, (vi) $y = x - \frac{x^2}{320}$ m, (vii) 80 m, (viii) 60 m

28. (i) 45°, (ii) $10\sqrt{2}$ m/s, (iii) 2 sec

29. (i) Bullet falls by $\frac{1}{2}gt^2$ which is exactly the amount of distance the monkey falls in this same time. So the bullet will hit the aim.

(ii) Acceleration of bullet w.r.t. monkey is zero then path of bullet will be appeared to monkey is straight line

(iii) If monkey does not drop from the tree then only bullet falls by $\frac{1}{2}gt^2$ so the bullet will miss the aim by distance $\frac{1}{2}gt^2$

30. (1) $20\sqrt{2}$ m/s (2) 75 m from the point of projection (3) 37.4 m/s (4) 20 m/s

31. 20 **32.** (3) **33.** 5 **34.** (4)

35. (2) **36.** (15 or 75) **37.** (2)

38. (i) $t = \frac{L_1 + L_2}{v_1 + v_2}$, (ii) $d_1 = v_1 \times \left(\frac{L_1 + L_2}{v_1 + v_2}\right), d_2 = v_2 \times \left(\frac{L_1 + L_2}{v_1 + v_2}\right)$

39. 50 km/h (60° N of E) **40.** 8 km/h, 30° S of W

41. 4 m/s In south direction, at 60° with vertical

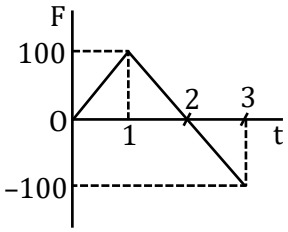
42. (1) **43.** (3) **44.** 3

Laws of Motion and Friction

1. A particle is acted upon by a force given by $F = (12t + 3t^2)N$, where t is in seconds. Find the change in momentum of that particle from t = 1 to t = 3 sec.

PLM257

2. For the given F-t graph. Find Impulse in first 3 sec.



PLM258

3. A ball of mass 2 kg is moving with velocity 20 m/sec strikes normally with the surface and rebounds with same speed, if time of contact is 0.1 sec then calculate average force on the ball.

PLM259

4. A machine gun fire bullets of mass 40 gm each with the velocity of 1000 m/sec. If average force is exerted on the gun F is 400 N, then find the no. of bullets fired per minute.

PLM260

5. When a same force is applied on two different objects. It produces accelerations of 4 m/sec^2 and 6 m/sec^2. In these objects, if the same force is applied on the combination of these objects, calculate acceleration of combination.

PLM261

6. Suppose a rocket with an initial mass M_0 eject a mass Δm in the form of gases in time Δt, then the mass of the rocket after time t is :-

(1) $M_0 - \frac{\Delta m}{\Delta t}\cdot t$ (2) $M_0 - \frac{\Delta m}{\Delta t}$

(3) $M_0 + \frac{\Delta m}{\Delta t}$ (4) M_0

PLM262

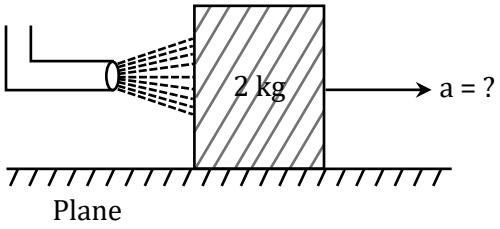
7. An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to : [Use $g = 10\ ms^{-2}$]

(1) 1 : 1 (2) $\sqrt{2}:\sqrt{3}$

(3) $\sqrt{3}:\sqrt{2}$ (4) 2 : 3

PLM263

8. A block of metal weighing 2 kg is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of 1 kgs^{-1} and at a speed of 10 ms^{-1}. Then, the initial acceleration of the block, in ms^{-2}, will be :



(1) 3 (2) 6 (3) 5 (4) 4

PLM264

9. A bag is gently dropped on a conveyor belt moving at a speed of 2 m/s. The coefficient of friction between the conveyor belt and bag is 0.4 Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion is : [Take $g = 10\ m/s^{-2}$]

(1) 2 m (2) 0.5 m

(3) 3.2 m (4) 0.8 ms

PLM265

10. In two different experiments, an object of mass 5 kg moving with a speed of 25 ms^{-1} hits two different walls and comes to rest within
(i) 3 second, (ii) 5 seconds, respectively.
Choose the correct option out of the following :

(1) Impulse and average force acting on the object will be same for both the cases.

(2) Impulse will be same for both the cases but the average force will be different.

(3) Average force will be same for both the cases but the impulse will be different.

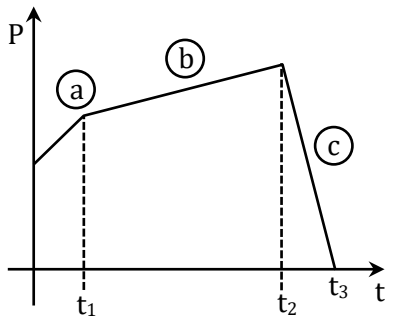
(4) Average force and impulse will be different for both the cases.

PLM266

11. A spherical body of mass 2 kg starting from rest acquires a kinetic energy of 10000 J at the end of 5th second. The force acted on the body is ___ N. (Assuming constant acceleration)

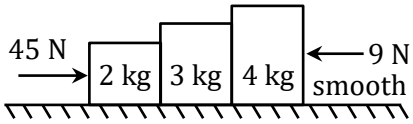
PLM267

12. The figure represents the momentum time (p-t) curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively? If $(t_3 - t_2) < t_1$.



(1) c and a (2) b and c
(3) c and b (4) a and b

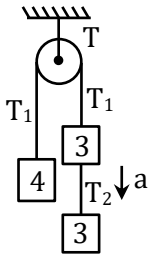
PLM268

13. 

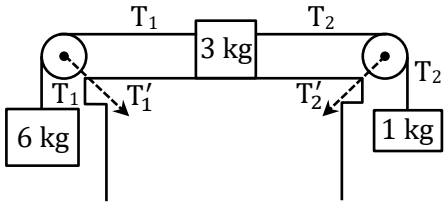
Find interaction force between 3 kg and 4 kg block.

PLM269

14. Find a, T_1, T_2, T'. (All masses are in Kg)
($g = 10\ m/s^2$)

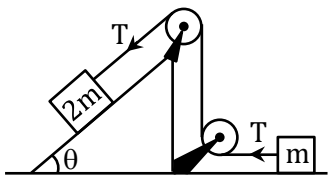


PLM270

15. 

Calculate a, T_1, T_2, T_1' & T_2'. ($g = 10\ m/s^2$)

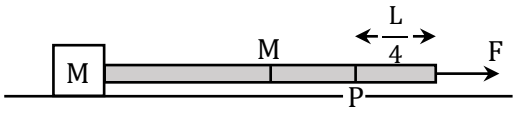
PLM271

16. 

Calculate a & T. ($g = 10\ m/s^2$)

PLM272

17. A block of mass M is pulled by a uniform chain of mass M tied to it by applying a force F at the other end of the chain. The tension force at a point located at a distance of one-fourth of the chain's length from the free end of chain will be: -



(1) $\frac{7F}{8}$ (2) $\frac{4F}{5}$
(3) $\frac{3F}{4}$ (4) $\frac{6F}{7}$

PLM273

18. A monkey of mass 50kg climbs on a rope which can withstand the tension (T) of 350N. If monkey initially climbs down with an acceleration of 4 m/s^2 and then climbs up with an acceleration of 5 m/s^2. Choose the correct option ($g = 10\ m/s^2$)

(1) T = 700N while climbing upward
(2) T = 350 N while going downward
(3) Rope will break while climbing upward
(4) Rope will break while going downward

PLM274

19. A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be :

(1) Mg
(2) Mg cos θ
(3) $\sqrt{Mg\sin\theta + Mg\cos\theta}$
(4) $Mg\sin\theta\sqrt{1+\mu}$

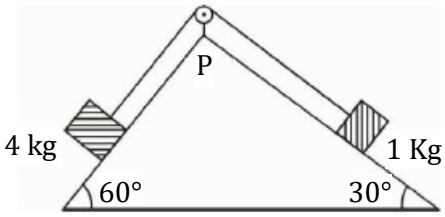
PLM275

20. A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

(1) 3 (2) 9 (3) 12 (4) 18

PLM276

21. As per given figure, a weightless pulley P is attached on a double inclined frictionless surface. The tension in the string (massless) will be (if g = 10 m/s²)



(1) $(4\sqrt{3}+1)$N (2) $4(\sqrt{3}+1)$N

(3) $4(\sqrt{3}-1)$N (4) $(4\sqrt{3}-1)$N

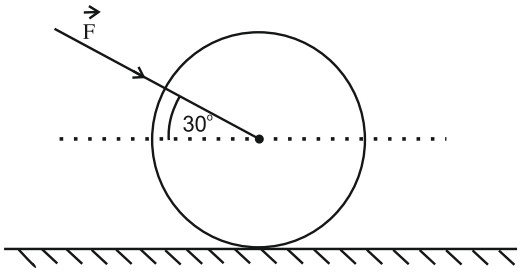
PLM277

22. Force acts for 20 s on a body of mass 20 kg, starting from rest, after which the force ceases and then body describes 50 m in the next 10 s. The value of force will be :

(1) 40 N (2) 5 N (3) 20 N (4) 10 N

PLM278

23. As shown in figure, a 70 kg garden roller is pushed with a force of $\vec{F}=200$N at an angle of 30° with horizontal. The normal reaction on the roller is (Given g = 10 m s^{-2})

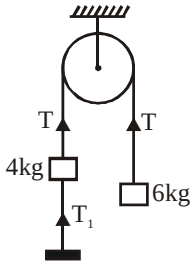


(1) $800\sqrt{2}$N (2) 600 N

(3) 800N (4) $200\sqrt{3}$N

PLM279

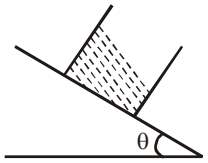
24. Two bodies of mass 4 kg and 6 kg are attached to the ends of a string passing over a pulley. The 4 kg mass is attached to the table top by other string. The tension in this string T_1 is equal to :-

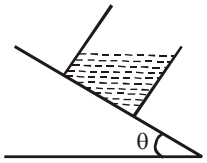


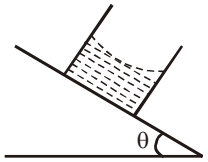
(1) 10 N (2) 10.6 N (3) 25 N (4) 20 N

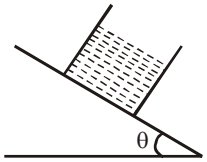
PLM280

25. A beaker is half filled with water. It is allowed to slip down on smooth inclined plane with angle of inclination θ to the horizontal. The level of water in the beaker will be: -

(1) 

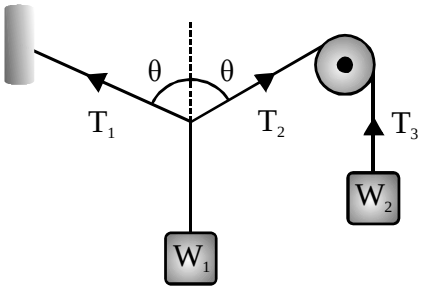
(2) 

(3) 

(4) 

PLM281

26. In the following figure, the pulley is massless and frictionless. The relation between T_1, T_2 and T_3 will be: -



(1) $T_1 = T_2 \neq T_3$ (2) $T_1 \neq T_2 = T_3$

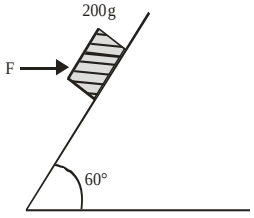
(3) $T_1 \neq T_2 \neq T_3$ (4) $T_1 = T_2 = T_3$

PLM282

27. A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _____m.

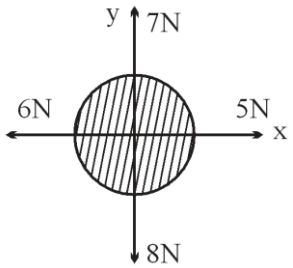
PLM283

28. A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F=\sqrt{x}$N as shown in figure. The value of x = _____.



PLM284

29. For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero?

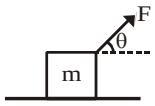


(1) $\sqrt{2}$ N, 45° (2) $\sqrt{2}$ N, 135°

(3) $\frac{2}{\sqrt{3}}$ N , 30° (4) 2N, 45°

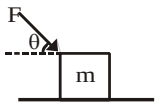
PLM285

30. Find value of pulling force F for which block just moves. Given coefficient of friction for surface is μ.



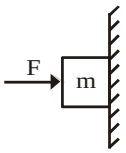
PLM286

31. Find value of pushing force F for which the body just moves. (Coefficient of friction for surface is μ)



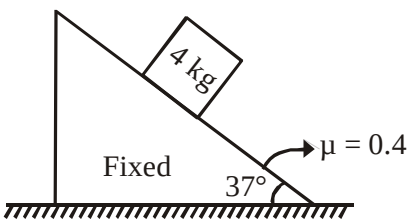
PLM287

32. Find the minimum value of horizontal force (F) required on the block of mass m to keep it at rest on the wall. Given the coefficient of friction between the surfaces is μ.



PLM288

33.



For the given arrangement, find :-

(i) Force of friction on the block.

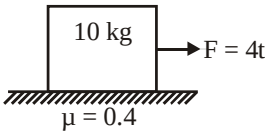
(ii) Acceleration of the block.

(iii) Minimum Force required to prevent the downward sliding of block.

(iv) Minimum Force required to just move the block upwards.

PLM289

34. A block of mass 10 kg is kept over a rough surface and a force F = 4t is applied on it. At what value of t the block will start moving :- ($g = 10\ m/s^2$)



(1) > 10 s

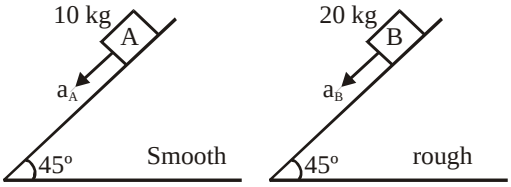
(2) < 8 s

(3) = 9 s

(4) None of these

PLM290

35.



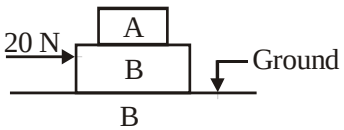
The ratio of acceleration of blocks A placed on smooth incline with block B placed on rough incline is 2:1. The coefficient of kinetic friction between block B and incline is :-

(1) 0.5 (2) 0.75

(3) 0.57 (4) None of these

PLM291

36.



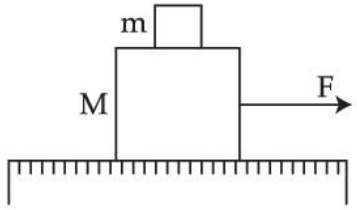
Blocks shown in figure moves with constant velocity 10 m/s towards right. All surfaces in contact are rough. The friction force applied by B on A is: -

(1) 0 N (2) 20 N

(3) 10 N (4) insufficient data

PLM292

37. A system of two blocks of masses m = 2 kg and M = 8 kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be :



(1) 9.8 N (2) 39.2 N
(3) 49 N (4) 78.4 N

PLM293

38. A block of mass 5 kg is placed at rest on a table of rough surface. Now, if a force of 30N is applied in the direction parallel to surface of the table, the block slides through a distance of 50 m in an interval of time 10s. Coefficient of kinetic friction is (given, g = 10 ms^{-2}):

(1) 0.60
(2) 0.75
(3) 0.50
(4) 0.25

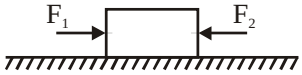
PLM294

ANSWER KEY

1.	74 kg ms^{-1}	**2.**	50 N-s	**3.**	800N	**4.**	600 bullets		
5.	a= 2.4 m/s^2	**6.**	(1)	**7.**	(2)	**8.**	(3)	**9.**	(2)
10.	(2)	**11.**	40	**12.**	(3)	**13.**	25 N		
14.	a = 2m/sec^2, T_1 = 48 N, T_2 = 24 N, T' = 96 N								
15.	$a = 5\ m/sec^2, T_1 = 30\ N, T_2 = 15\ N, T_1' = 30\sqrt{2}\ N,\ T_2' = 15\sqrt{2}\ N$								
16.	$a = \frac{2g\sin\theta}{3}, T = \frac{2mg\sin\theta}{3}$			**17.**	(1)	**18.**	(3)	**19.**	(1)
20.	(2)	**21.**	(2)	**22.**	(2)				
23.	(3)	**24.**	(4)	**25.**	(4)	**26.**	(4)	**27.**	2
28.	12	**29.**	(1)	**30.**	$F = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$				
31.	$F = \frac{\mu mg}{\cos\theta - \mu\sin\theta}$			**32.**	$F = \frac{mg}{\mu}$				
33.	(i) 12.8 N, (ii) 2.8 m/s^2, (iii) 11.2 N, (iv) 36.8 N								
34.	(1)	**35.**	(1)	**36.**	(1)	**37.**	(3)	**38.**	(3)

Work, Energy and Power

1. A block of mass 2 kg is placed on a smooth horizontal surface. Two forces F_1 = 20 N and F_2 = 5 N start acting on the block in opposite directions as shown. If block gets displaced by 5 m in the direction of net force then work done by F_2 is :-



(1) – 75 J (2) 75 J
(3) – 25 J (4) 25 J

PWP156

2. A man of mass 50 kg is standing in an elevator. If elevator is moving up with an acceleration $\frac{g}{3}$ then work done by normal reaction of elevator floor on man when elevator moves by a distance 12 m is ($g = 10\ m/s^2$) :-

(1) 2000 J (2) 4000 J
(3) 6000 J (4) 8000 J

PWP157

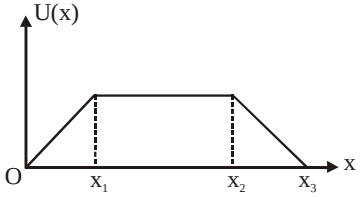
3. A man pulls a bucket full of water from h metre deep well. If the mass of rope is m and mass of bucket full of water is M, then work done by the man is :-

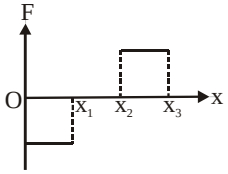
(1) $\left(\frac{M}{2}+m\right)gh$ (2) $\left(\frac{M+m}{2}\right)gh$

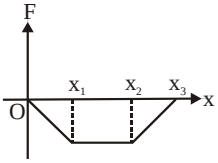
(3) $\left(M+\frac{m}{2}\right)gh$ (4) (M + m) gh

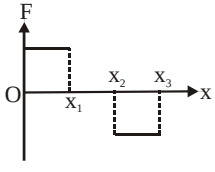
PWP158

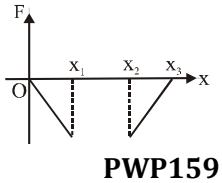
4. The variation of potential energy U of a system is shown in figure. The force acting on the system is best represented by :-



(1)


(2)


(3)


(4)


PWP159

5. A particle experiences a variable force $\vec{F}=(4x\hat{i}+3y^2\hat{j})$ in a horizontal x-y plane. Assume distance in meters and force is newton. If the particle moves from point (1, 2) to point (2, 3) in the x-y plane, the Kinetic Energy changes by

(1) 50.0 J (2) 12.5 J (3) 25.0 J (4) 0 J

PWP160

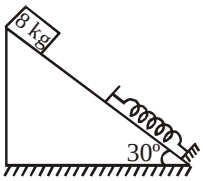
6. Potential energy as a function of r is given by $U=\frac{A}{r^{10}}-\frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be :

(1) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$ (2) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$

(3) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$ (4) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

PWP161

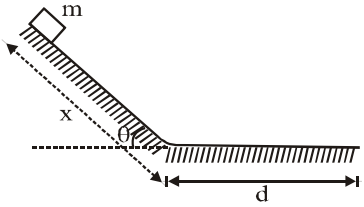
7. A block of mass 8 kg is released from the top of an inclined smooth surface as shown in figure. If spring constant of spring is 200 N/m and block comes to rest after compressing spring by 1 m then find the distance travelled by block before it comes to rest:-($g = 10\ m/s^2$)



(1) 2.5 m (2) 3.5 m
(3) 2.0 m (4) None of these

PWP162

8. A block of mass m is released from the top of a smooth inclined plane of length x and inclination θ as shown in figure. Horizontal surface is rough. If block comes to rest after moving a distance d on the horizontal surface, then coefficient of friction between block and surface is: -

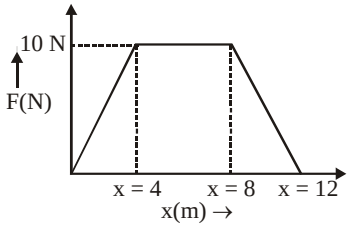


(1) $\frac{x \sin \theta}{2d}$ (2) $\frac{x \cos \theta}{2d}$

(3) $\frac{x \sin \theta}{d}$ (4) $\frac{x \cos \theta}{d}$

PWP163

9. A particle of mass 0.1 kg is subjected to a force which varies with distance as shown. If it starts its journey from rest at x = 0, then its velocity at x = 12 m is: -



(1) 0 m/s (2) $20\sqrt{2}$ m/s

(3) $20\sqrt{3}$ m/s (4) 40 m/s

PWP164

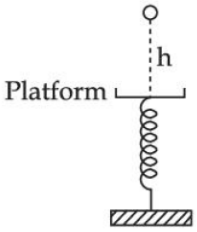
10. The rate of doing work by force acting on a particle moving along x-axis depends on position x of particle and is equal to 2x. The velocity of particle is given by expression: -

(1) $\left[\frac{3x^2}{m}\right]^{1/3}$ (2) $\left[\frac{3x^2}{2m}\right]^{1/3}$

(3) $\left[\frac{2mx}{9}\right]^{1/2}$ (4) $\left[\frac{mx^2}{3}\right]^{1/2}$

PWP165

11. A ball of mass 100 g is dropped from a height h = 10 cm on a platform fixed at the top of vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance $\frac{h}{2}$. The spring constant is _____ Nm^{-1}. (Use g = 10 ms^{-2})



PWP166

12. As per the given figure, two blocks each of mass 250g are connected to a spring of spring constant $2Nm^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is :



(1) $\frac{v}{2\sqrt{2}}$ (2) $\frac{v}{2}$

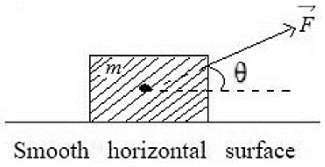
(3) $\frac{v}{4}$ (4) $\frac{v}{\sqrt{2}}$

PWP167

13. A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = \left(t\hat{i} + 3t^2\hat{j}\right)N$. where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y axis. The power developed by above force, at the time t = 2s. will be _____W.

PWP168

14. An object of mass 'm' initially at rest on a smooth horizontal plane starts moving under the action of force F = 2N. In the process of its linear motion, the angle θ (as shown in figure) between the direction of force and horizontal varies as θ = kx, where k is a constant and x is the distance covered by the object from its initial position. The expression of kinetic energy of the object will be $E = \frac{n}{k}\sin\theta$. The value of n is___.



PWP169

15. A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in 4s is $\frac{1}{3}\alpha^2\sqrt{P}$ m. The value of α will be .....

PWP170

ANSWER KEY

1.	(3)	**2.**	(4)	**3.**	(3)	**4.**	(1)	**5.**	(3)
6.	(3)	**7.**	(1)	**8.**	(3)	**9.**	(4)	**10.**	(1)
11.	120	**12.**	(2)	**13.**	100	**14.**	2	**15.**	4

Circular Motion

1. If a_r and a_t represents radial and tangential acceleration, the motion of a particle will be circular if :-

(i) $a_r = 0$ and $a_t = 0$ (ii) $a_r = 0$ but $a_t \neq 0$

(iii) $a_r \neq 0$ but $a_t = 0$ (iv) $a_r \neq 0$ and $a_t \neq 0$

(1) (i, iii) (2) (ii, iii)

(3) (iii, iv) (4) (ii, iv)

PCM115

2. A particle P is moving in a circle of radius 'a' with uniform speed v. C is the centre of the circle and AB is a diameter. The angular velocity of P about A and C are in the ratio :-

(1) 1 : 1 (2) 1 : 2

(3) 2 : 1 (4) 4 : 1

PCM116

3. A motor car is travelling at 60 m/s on a circular road of radius 1200 m. Its speed is increasing at the rate of 4 m/s^2. The acceleration of the car is :-

(1) 3 m/s^2 (2) 4 m/s^2

(3) 5 m/s^2 (4) 7 m/s^2

PCM117

4. The speed of a particle moving in a circle slows down at a rate of 3 m/sec^2. At some instant the magnitude of the total acceleration is 5 m/sec^2 and the particle's speed is 12 m/sec. The radius of circle will be :

(1) 12 m (2) 24 m (3) 36 m (4) 48 m

PCM118

5. A particle moves along a circle with speed v = at. The total acceleration of the particle at a time when it has traced 1/8th of the circumference is :-

(1) $\frac{v}{8a}$ (2) $2a\sqrt{4+\pi^2}$

(3) a (4) $\frac{a}{2}\sqrt{4+\pi^2}$

PCM119

6. Railway tracks are banked at the curves so that :-

(1) the train may not fall down inwards

(2) the weight of the train may be reduced

(3) the necessary centripetal force to train may be obtained from the horizontal component of the reaction force of railway tracks

(4) no frictional force may be produced between the wheel and the track

PCM120

7. A spaceman during training period is rotated in a seat at the end of a horizontal arm of length 5 m. If he can withstand acceleration upto 9 g, then what is the maximum number of revolutions per second permissible?

(Take g = 10 m/s^2)

(1) 13.5 rev/s (2) 1.35 rev/s

(3) 0.675 rev/s (4) 6.75 rev/s

PCM121

8. The maximum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is :- (g = 10 m/s^2)

(1) 60 (2) 30 (3) 15 (4) 25

PCM122

9. A 2 kg stone at the end of a string 1 m long is whirled in a vertical circle. At some point its speed is 4m/s. The tension of the string is 51.6 newton. At this instant the stone is :-

(1) at the top of the circle

(2) at the bottom of the circle

(3) half way down

(4) none of these

PCM123

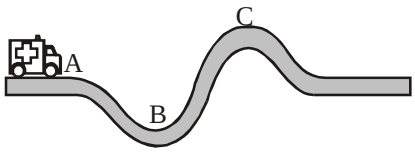
10. A particle rests on the top of a hemisphere of radius R. Find the smallest horizontal velocity that must be imparted to the particle if it is to leave the hemisphere without sliding down :-

(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$

(3) $\sqrt{3gR}$ (4) $\sqrt{5gR}$

PCM124

11. A vehicle is moving on a track with constant speed as shown in Fig. The normal reaction between tyre's and road is :-



(1) maximum at A (2) maximum at B

(3) maximum at C (4) same at A,B and C

PCM125

12. A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is

(1) 3 (2) 2 (3) 1 (4) 5

PCM126

13. A body is moving with constant speed, in a circle of radius 10 m. The body completes one revolution in 4 s. At the end of 3rd second, the displacement of body (in m) from its starting point is:

(1) 30

(2) 15π

(3) 5π

(4) $10\sqrt{2}$

PCM127

14. A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x} ms^{-2}$. The value of x _____.

$\left(\text{Take } \pi = \frac{22}{7}\right)$

PCM128

ANSWER KEY

1.	(3)	**2.**	(2)	**3.**	(3)	**4.**	(3)	**5.**	(4)
6.	(3)	**7.**	(3)	**8.**	(2)	**9.**	(2)	**10.**	(1)
11.	(2)	**12.**	(2)	**13.**	(4)	**14.**	125		

Collisions and Centre of Mass

1. A disc of radius r is cut from a larger disc of radius 4r in such a way that the edge of the hole touches the edge of the disc. The centre of mass of the residual disc will be at a distance from centre of larger disc :-

(1) $\frac{r}{5}$ (2) $\frac{r}{4}$ (3) $\frac{r}{2}$ (4) $\frac{r}{3}$

PSC126

2. Two particles of mass m_1 and m_2 are projected from the top of a tower. The particle m_1 is projected vertically downward with speed u and m_2 is projected horizontally with same speed. Find acceleration of CM of system of particles by neglecting the effect of air resistance.

(1) g downward

(2) $\frac{m_1 g}{m_1 + m_2}$ downward

(3) $\frac{m_2 g}{m_1 + m_2}$ downward

(4) Can't be predicated

PSC127

3. A cart of mass M is tied to one end of a massless rope of length 10m. The other end of the rope is in the hands of a man of mass $\frac{M}{2}$. The entire system is on a smooth horizontal surface. If the man pulls the cart by the rope then find the distance by which the man will slip on the horizontal surface before the man and the cart will meet each other.

(1) $\frac{20}{3}$m (2) $\frac{10}{3}$m

(3) 0 (4) None of these

PSC128

4. Two point masses m_1 and m_2 at rest in gravity free space are released from distance d. Find the velocity of the CM of the system at the time of collision of particles.

(1) Zero (2) 2 (3) 3 (4) None

PSC129

5. Two blocks of masses 5kg and 2kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse provides a velocity of 7m/s to the heavier block in the direction of the lighter block. The velocity of the centre of mass is :-

(1)4 m/s (2) 5m/s (3) 2m/s (4) 3m/s

PSC130

6. A shell at rest on a smooth horizontal surface explodes into two fragments of masses m_1 and m_2. If just after explosion, m_1 move with speed u, then work done by internal forces during explosion is:-

(1) $\frac{1}{2}(m_1 + m_2)\frac{m_2}{m_1}u^2$ (2) $\frac{1}{2}(m_1 + m_2)u^2$

(3) $\frac{1}{2}m_1u^2\left(1+\frac{m_1}{m_2}\right)$ (4) $\frac{1}{2}(m_2 - m_1)u^2$

PSC131

7. A body of mass 2 kg moving with a velocity of 3 m/s collides head on with a body of mass 1 kg moving with a velocity of 4 m/s in opposite direction. After collision the two bodies stick together and move with a common velocity :-

(1) (1/4) m/s (2) (1/3) m/s
(3) (2/3) m/s (4) (3/4) m/s

PSC132

8. A ball is dropped on the ground from a height of 1m. the coefficient of restitution is 0.6. The height to which the ball will rebound is :-

(1) 0.6 m (2) 0.4 m
(3) 0.36 m (4) 0.16 m

PSC133

9. Two billiard balls of mass 0.05 kg each moving in opposite directions with 10 ms^{-1} collide and rebound with the same speed. If the time duration of contact is t = 0.005 s, then what is the force exerted on the ball due to each other?

(1) 100 N (2) 200 N
(3) 300 N (4) 400 N

PSC134

10. A ball is dropped from a height of 20 m. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of_____m.

PSC135

ANSWER KEY

1.	(1)	**2.**	(1)	**3.**	(1)	**4.**	(1)	**5.**	(2)
6.	(3)	**7.**	(3)	**8.**	(3)	**9.**	(2)	**10.**	(5)

Rotational Motion

1. A fan is rotating with a speed of 450 rev/minute. After being switched off it comes to rest in 10s. Assuming constant angular deceleration, calculate the number of revolutions made by it before coming to rest.

(1) 37.5 (2) 37

(3) 38.5 (4) None of these

PRM207

2. A disc rotating about its axis from rest, acquires the angular speed 100 rev/s in 4 second. The angle rotated by it during these seconds (in radian) is :-

(1) 100 π (2) 200 π

(3) 300 π (4) 400 π

PRM208

3. A body rotating with uniform angular acceleration covers 100π (radian) in the first 5s after the start. Its angular speed at the end of 5s (in radian/s) is :-

(1) 40 π (2) 30 π

(3) 20 π (4) 10 π

PRM209

4. A wheel starting from rest is uniformly accelerated at 2 rad/s² for 20 seconds. It is allowed to rotate uniformly for the next 10 seconds and is finally brought to rest in next 20 seconds. The total angle rotated by the wheel (in radian) is :-

(1) 600 (2) 1200

(3) 1800 (4) 300

PRM210

5. A body rotates about a fixed axis with an angular acceleration of 3 rad/s². The angle rotated by it during the time when its angular velocity increases from 10 rad/s to 20 rad/s (in radian) is :-

(1) 50 (2) 100 (3) 150 (4) 200

PRM211

6. For a disc of given density and thickness, its moment of inertia varies with radius of the disc as :-

(1) $I \propto R^2$ (2) $I \propto R^4$

(3) $I \propto R^3$ (4) $I \propto R$

PRM212

7. If a disc of moment of inertia I and radius r is reshaped into a ring of radius nr, keeping its mass same, its moment of inertia becomes :-

(1) n^2I (2) $2n^2I$ (3) $\frac{n^2}{2}I$ (4) I

PRM213

8. Two circular loops A and B of radii R and 2R respectively are made of the similar wire. Their moment of inertia about the axis passing through the centre and perpendicular to their plane are I_A and I_B respectively. The ratio $\frac{I_A}{I_B}$ is :-

(1) $\frac{1}{4}$ (2) $\frac{1}{8}$ (3) $\frac{1}{2}$ (4) 1

PRM214

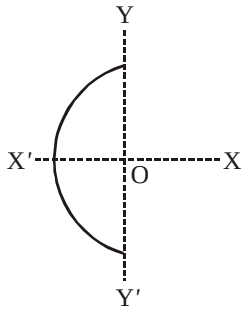
9. The radius of gyration (K) of a rigid body changes with change of :-

(1) Angular speed (2) Axis of rotation

(3) Both (1) and (2) (4) Never changes

PRM215

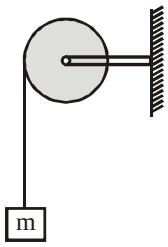
10. A thin wire of length ℓ and mass m is bent in the form of a semicircle as shown. Its moment of inertia about an axis joining its free ends will be :-



(1) $m\ell^2$ (2) Zero (3) $\frac{m\ell^2}{\pi^2}$ (4) $\frac{m\ell^2}{2\pi^2}$

PRM216

11. A disc of mass m and radius r is free to rotate about its centre as shown in the figure. A string is wrapped over its rim and a block of mass m is attached to the free end of the string. The system is released from rest. The speed of the block as it descends through a height h, is :-



(1) $\sqrt{2gh}$ (2) $\sqrt{\frac{2}{3}gh}$

(3) $2\sqrt{\frac{gh}{3}}$ (4) $\frac{1}{2}\sqrt{3gh}$

PRM217

12. A particle of mass m has been thrown up with initial speed u making an angle θ with the horizontal. Find the torque of its weight about the point of projection when it just reached the highest point.

(1) mu sinθ·cosθ (2) m u^2 sin θ cos θ

(3) $\frac{mu\sin^2\theta}{2}$ (4) $\frac{mu^2\sin^2\theta}{2}$

PRM218

13. The torque of force $\vec{F}=-2\hat{i}+2\hat{j}+3\hat{k}$ acting on a point $\vec{r}=\hat{i}-2\hat{j}+\hat{k}$ about origin will be :-

(1) $8\hat{i}+5\hat{j}+2\hat{k}$ (2) $-8\hat{i}-5\hat{j}-2\hat{k}$

(3) $8\hat{i}-5\hat{j}+2\hat{k}$ (4) $-8\hat{i}+5\hat{j}-2\hat{k}$

PRM219

14. Moment of a force of magnitude 20 N acting along positive x direction at point (3m, 0, 0) about the point (0, 2m, 0) (in N-m) is :-

(1) 20 (2) 60 (3) 40 (4) 30

PRM220

15. A disc is rotating with angular velocity $\vec{\omega}$. A force $\vec{F}$ acts at a point whose position vector with respect to the axis of rotation is $\vec{r}$. The power associated with torque due to the force is given by :

(1) $(\vec{r}\times\vec{F})\cdot\vec{\omega}$ (2) $(\vec{r}\times\vec{F})\times\vec{\omega}$

(3) $\vec{r}\times(\vec{F}\cdot\vec{\omega})$ (4) $\vec{r}\cdot(\vec{F}\times\vec{\omega})$

PRM221

16. When torque applied on a system is zero, which of the following will be constant ?

(1) Moment of inertia

(2) Angular velocity

(3) Kinetic energy

(4) Moment of linear momentum

PRM222

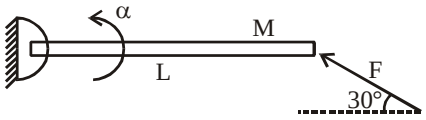
17. Two boys are holding a horizontal rod of length L and weight W through its two ends. If now one of the boys suddenly leaves the rod, what is the instantaneous reaction force experienced by the other boy ?

(1) $\frac{W}{4}$ (2) $\frac{W}{2}$

(3) $\frac{3W}{4}$ (4) W

PRM223

18. The thin rod shown below has mass M and length L. A force F acts at one end as shown and the rod is free to rotate about the other end in horizontal plane. Initial angular acceleration of the rod is :

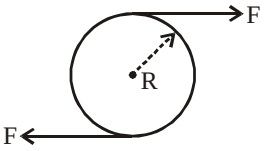


(1) $\frac{3F}{2ML}$ (2) $\frac{2F}{3ML}$

(3) $\frac{F}{ML}$ (4) $\frac{F}{2ML}$

PRM224

19. Two equal and opposite forces are applied tangentially on a uniform disc of mass M and radius R as shown in the figure. If the disc is pivoted at its centre and free to rotate in its plane, the angular acceleration of the disc is :

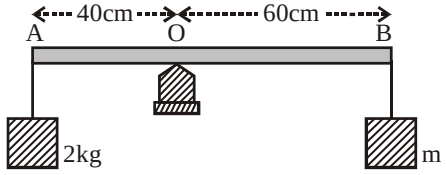


(1) $\frac{F}{MR}$ (2) $\frac{2F}{3MR}$

(3) $\frac{4F}{MR}$ (4) Zero

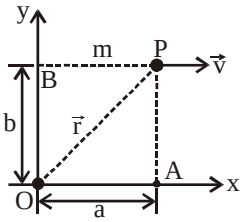
PRM225

20. Figure shows a rigid massless rod of length 1.0 m. It is pivoted at O. For what value m, the rod will be in equilibrium ? Find the force (F) exerted on the rod by the pivot. Neglect the weight of the rod. ($g = 10\ m/s^2$)



PRM226

21. A particle is moving along a straight line parallel to x-axis with constant velocity. Find angular momentum about the origin in vector form :-

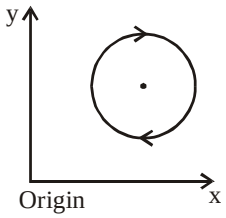


(1) $+ mv^2b\,\hat{k}$ (2) $- mvb\,\hat{k}$

(3) $-2mvb\,\hat{k}$ (4) $-mvb\,\hat{j}$

PRM227

22. A particle is moving in circle with uniform speed as shown. The angular momentum of the particle w.r.t. origin is :-



(1) Constant in magnitude only

(2) Constant in magnitude as well as direction

(3) Constant in direction only

(4) Variable in magnitude as well as direction

PRM228

23. A particle is moving along a straight line with increasing speed. Its angular momentum about a fixed point on this line :-

(1) Goes on increasing

(2) Goes on decreasing

(3) May be increasing or decreasing depending on direction of motion

(4) Remains zero

PRM229

24. If radius of earth suddenly contracts by half of its initial value keeping mass constant, then what will be the time period of rotation of earth about it's axis ?

(1) 6hr (2) 12hr (3) 3hr (4) 4hr

PRM230

25. Due to global warming, ice on polar caps is likely to melt in large quantity. Due to this effect :-

(1) Moment of inertia of earth shall decrease

(2) Duration of the day shall decrease

(3) Angular velocity of earth shall decrease

(4) Angular momentum of earth shall decrease

PRM231

26. A disc of mass 1 kg and radius 0.1 m is rotating with angular velocity 20 rad/s. What is angular velocity (in rad/s) if a mass of 0.5 kg is put on the circumference of the disc ?

(1) 10 (2) 20 (3) 40 (4) 30

PRM232

27. A metre stick is pivoted about its centre. A piece of wax of mass 20g travelling horizontally and perpendicular to it at 5 m/s strikes and adheres to one end of the stick so that the stick starts to rotate in a horizontal circle. Given the moment of inertia of the stick and wax about the pivot is 0.02 kg m^2, the initial angular velocity of the stick is :-

(1) 1.58 rad/s (2) 2.24 rad/s

(3) 2.50 rad/s (4) 5.00 rad/s

PRM233

28. If two discs of moment of inertia I_1 and I_2 rotating about collinear axis passing through their centre of mass and perpendicular to their plane with angular speeds, ω_1 and ω_2 respectively in opposite directions. Now they are made to rotate combinedly along same axis, then the magnitude of angular velocity of the system is :-

(1) $\frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$ (2) $\frac{I_1\omega_1 - I_2\omega_2}{I_1 + I_2}$

(3) $\frac{I_1\omega_1 + I_2\omega_2}{\omega_1 + \omega_2}$ (4) $\frac{I_1\omega_1 - I_2\omega_2}{\omega_1 - \omega_2}$

PRM234

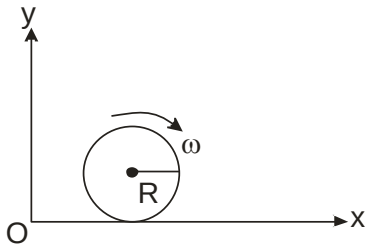
29. An angular impulse of 20 Nms is applied to a hollow cylinder of mass 2 kg and radius 20 cm. The change in its angular speed is :-

(1) 25 rad/s (2) 2.5 rad/s

(3) 250 rad/s (4) 2500 rad/s

PRM235

30. A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3}R^2\omega$. The value of a will be :



(1) 2 (2) 3 (3) 5 (4) 4

PRM236

31. Choose the correct statement.

(1) A sphere can do pure rolling on smooth horizontal surface

(2) A sphere can't do pure rolling on a fixed smooth wedge

(3) Friction can act parallel or antiparallel to the direction of motion

(4) All of these

PRM237

32. Two solid spheres of different mass, radii and density roll down a rough inclined plane under identical situation. Their time to come down is independent of their :-

(1) Mass (2) Radius

(3) Density (4) All of these

PRM238

33. When a body is under pure rolling, the fraction of its total kinetic energy which is purely rotational is $\frac{2}{5}$. Identify the body.

(1) Hollow sphere (2) Sphere

(3) Ring (4) Disc

PRM239

34. A solid sphere rolls on horizontal surface without slipping. What is the ratio of its rotational to translation kinetic energy.

(1) $\frac{2}{5}$ (2) $\frac{5}{2}$

(3) $\frac{3}{2}$ (4) 0

PRM240

35. Two discs having masses in the ratio 1 : 2 and radii in the ratio 1 : 8 roll down without slipping one by one from an inclined plane of height h. The ratio of their linear velocities on reaching the ground is :-

(1) 1 : 16 (2) 1 : 128

(3) $1 : 8\sqrt{2}$ (4) 1 : 1

PRM241

36. Which of the following (if mass and radius are assumed to be same) have maximum percentage of total K.E. in rotational form while pure rolling?

(1) Disc (2) Sphere

(3) Ring (4) Hollow sphere

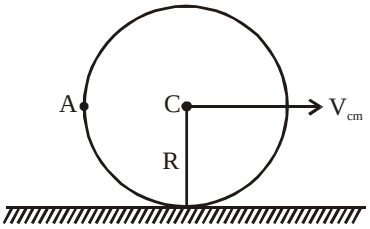
PRM242

37. A solid sphere of diameter 0.1 m and mass 5 kg is rolling down an inclined plane with a speed of 4 m/s. The total kinetic energy of the sphere is :-

(1) 28 J (2) 56 J (3) 84 J (4) 112 J

PRM243

38. In case of pure rolling, what will be the velocity of point A of the ring of radius R?



(1) V_{cm} (2) $\sqrt{2}V_{cm}$

(3) $\frac{V_{cm}}{2}$ (4) $2V_{cm}$

PRM244

39. When a body is rolling without slipping on a rough horizontal surface, the work done by friction is :-

(1) Always zero

(2) May be zero

(3) Always positive

(4) Always negative

PRM245

40. An inclined plane makes an angle of 30° with the horizontal. A solid sphere rolling down on this inclined plane from rest without slipping has a linear acceleration equal to :-

(1) $\frac{g}{3}$ (2) $\frac{2g}{3}$

(3) $\frac{5g}{7}$ (4) $\frac{5g}{14}$

PRM246

ANSWER KEY

1.	(1)	**2.**	(4)	**3.**	(1)	**4.**	(2)	**5.**	(1)
6.	(2)	**7.**	(2)	**8.**	(2)	**9.**	(2)	**10.**	(4)
11.	(3)	**12.**	(2)	**13.**	(2)	**14.**	(3)	**15.**	(1)
16.	(4)	**17.**	(1)	**18.**	(1)	**19.**	(3)		
20.	(m = 4/3 kg, F = 100/3 N)			**21.**	(2)	**22.**	(4)	**23.**	(4)
24.	(1)	**25.**	(3)	**26.**	(1)	**27.**	(3)	**28.**	(2)
29.	(3)	**30.**	(3)	**31.**	(4)	**32.**	(4)	**33.**	(1)
34.	(1)	**35.**	(4)	**36.**	(3)	**37.**	(2)	**38.**	(2)
39.	(1)	**40.**	(4)						

Gravitation

1. Two particles of equal mass m go round a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is

(1) $\frac{1}{2R}\sqrt{\frac{1}{Gm}}$ (2) $\sqrt{\frac{Gm}{2R}}$

(3) $\frac{1}{2}\sqrt{\frac{Gm}{R}}$ (4) $\sqrt{\frac{4Gm}{R}}$

PGR186

2. A particle is projected vertically upwards from the surface of the earth (radius R_e) with a speed equal to one fourth of escape velocity. What is the maximum height attained by it from the surface of the earth ?

(1) $\frac{16}{15}R_e$ (2) $\frac{R_e}{15}$ (3) $\frac{4}{15}R_e$ (4) None

PGR187

3. Which of the following statements are true about acceleration due to gravity?

(a) 'g' decreases when moving away from the centre if r > R

(b) 'g' decreases when moving away from the centre if r < R

(c) 'g' is zero at the central of earth

(d) 'g' decreases if earth stops rotating about its axis

(1) (a) & (b) (2) (a), (b) & (c)

(3) (a) & (c) (4) (a), (b) (c) & (d)

PGR188

4. The intensity of gravitational field at a point situated at a distance 8000 km from the centre of Earth is 6.0 newton /kg. The gravitational potential at that point in newton – meter/kg will be :-

(1) – 6 (2) -4.8×10^7

(3) -8×10^5 (4) -4.8×10^2

PGR189

5. If suddenly the gravitational force of attraction between earth and a satellite revolving around it becomes zero, then the satellite will-

(1) continue to move in its orbit with same speed

(2) move tangentially to the original orbit with same velocity

(3) become stationary in its orbit

(4) move towards the earth

PGR190

6. The kinetic energy needed to project a body of mass m from the earth's surface (radius R) to infinity is-

(1) $\frac{mgR}{2}$ (2) 2mgR

(3) mgR (4) $\frac{mgR}{4}$

PGR191

7. The height at which the acceleration due to gravity becomes g/9 (where g = acceleration due to gravity on the surface of the earth) in terms of R, (the radius of the earth) is :-

(1) $\frac{R}{2}$ (2) $\sqrt{2}R$ (3) 2R (4) $\frac{R}{\sqrt{2}}$

PGR192

8. Two bodies of masses m and 4m are placed at a distance r. The gravitational potential at a point on the line joining them where the gravitational field is zero is :-

(1) $-\frac{6Gm}{r}$ (2) $-\frac{9Gm}{r}$

(3) zero (4) $-\frac{4Gm}{r}$

PGR193

9. Gravitational force between two point masses m and M separated by a distance is F. Now if a point mass 2m is placed next to m in contact with it, the force on M due to m and the total force on M are :-

(1) 2F, F (2) F, 2F (3) F, 3F (4) F, F

PGR194

10. The height at which the acceleration due to gravity decreases by 36% of its value on the surface of the earth is: (Assume radius of the earth is R) :-

(1) $\frac{R}{4}$ (2) $\frac{R}{2}$

(3) $\frac{R}{6}$ (4) 4 R

PGR195

11. When a body is taken from pole to the equator its weight :-

(1) Remains constant

(2) Increases

(3) Decreases

(4) None of these

PGR196

12. The change in potential energy when a body of mass m is raised to a height nR from the earth's surface is: (R = radius of the earth)

(1) $mgR\frac{n}{(n-1)}$ (2) mgR

(3) $mgR\frac{n}{(n+1)}$ (4) $mgR\frac{n^2}{(n^2+1)}$

PGR197

13. The variation in the speed of the planet in its orbit about the sun can be explained on the basis of the conservation of :-

(1) Angular kinetic energy

(2) Linear momentum

(3) Angular momentum

(4) None of these

PGR198

ANSWER KEY

1.	(3)	**2.**	(2)	**3.**	(3)	**4.**	(2)	**5.**	(2)
6.	(3)	**7.**	(3)	**8.**	(2)	**9.**	(3)	**10.**	(1)
11.	(3)	**12.**	(3)	**13.**	(3)				

Properties of Matter and Fluid Mechanics-1

1. Blood is flowing at the rate of 200 cm^3/sec in a capillary of cross-sectional area 0.5m^2. The velocity of flow, (in mm/sec) is :-
(1) 0.1 (2) 0.2 (3) 0.3 (4) 0.4
PPF209

2. The cylindrical tube of a spray pump has a cross-section of 8cm^2, one end of which has 40 fine holes each of area 10^{-8} m^2. If the liquid flows inside the tube with a speed of 0.15m min^{-1}, the speed with which the liquid is ejected through the holes is :-
(1) 50 ms^{-1} (2) 5 ms^{-1}
(3) 0.05 ms^{-1} (4) 0.5 ms^{-1}
PPF210

3. A cylindrical vessel is filled with water upto a height of 92 cm. It has four holes 1,2,3 and 4 which are respectively at heights of 20cm, 30cm, 46cm and 80cm from the horizontal floor. The water falling at the maximum horizontal distance from the vessel comes from :-
(1) hole 4 (2) hole 3
(3) hole 2 (4) hole 1
PPF211

4. A cylindrical tank has a hole of 1 cm^2 in its bottom. If the water is allowed to flow into the tank from a tubes above it at the rate of 70 cm^3/sec then the maximum height upto which water can rise in the tank is :-
(1) 2.5 cm (2) 5 cm
(3) 10 cm (4) 0.25 cm
PPF212

5. An air bubble of 1 cm radius is rising at a steady rate of 2.00 mm/sec through a liquid of density 1.5 gm per cm^3. Neglect density of air. If g is 1000 cm/sec^2, then the coefficient of viscosity of the liquid is :-
(1) 0.166×10^3 poise (2) 166×10^3 poise
(3) 1.66×10^3 poise (4) 16.6×10^3 poise
PPF213

6. Sixty four spherical rain drops of equal size are falling vertically through air with a terminal velocity 1.5 ms^{-1}. If these drops coalesce to form a big spherical drop, then terminal velocity of big drop is :-
(1) 8 ms^{-1} (2) 16 ms^{-1}
(3) 24 ms^{-1} (4) 32 ms^{-1}
PPF214

7. A layer of glycerine of thickness 1 mm is present between a large surface and small surface of area 0.1m^2. With what force the small surface is to be pulled, so that it can move with a velocity of 1 m/s?
(coefficient of viscosity = 0.07kg-$m^{-1}s^{-1}$)
(1) 70N (2) 7N
(3) 700 N (4) 0.70N
PPF215

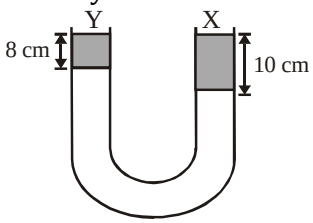
8. Water contained in a tank flows through an orifice of diameter 2cm, under a constant pressure difference of 10 cm of water column. The rate of flow of water through the orifice is nearly equal to :-
(1) 44 cc/sec (2) 4.4 cc/sec
(3) 440 cc/sec (4) 4400 cc/sec
PPF216

9. A sphere of brass released in a long liquid column attains a terminal velocity v_0. If the terminal velocity attained by sphere of marble of the same radius and released in the same liquid is nv_0. Then the value of n will be: (given the specific gravity of brass, marble and liquid are 8.5, 2.5 and 0.8 respectively.)
(1) $\frac{5}{17}$ (2) $\frac{17}{77}$ (3) $\frac{11}{31}$ (4) $\frac{17}{5}$
PPF217

10. Water flows in a stream-line manner through a capillary tube of radius a. The pressure difference being P and the rate of flow is Q. If the radius is reduced to a/2 and the pressure is increased to 2P, then the rate of flow becomes:-
(1) 4Q (2) 2Q
(3) Q (4) Q/8
PPF218

11. A tank of height 5m is half filled with water then it is filled upto top with a liquid of density 0.85 g/cm^3 then total pressure at bottom due to the liquids will be –
(1) 1.85 $\frac{\text{g-wt}}{\text{cm}^2}$ (2) 89.25 $\frac{\text{g-wt}}{\text{cm}^2}$
(3) 462.5 $\frac{\text{g-wt}}{\text{cm}^2}$ (4) 500 $\frac{\text{g-wt}}{\text{cm}^2}$
PPF219

12. A liquid X of density 3.36 g/cm^3 is poured in a U-tube, which contains Hg. Another liquid Y is poured in left arm with height 8 cm. Upper levels of X and Y are same.
(Density of Hg = 13.6 × 10^3 kg/m^3)
What is density of Y?



(1) 0.8 gm/cc (2) 1.2 gm/cc
(3) 1.4 gm/cc (4) 1.6 gm/cc

PPF220

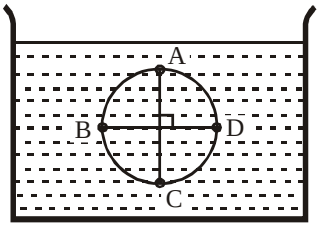
13. A body of uniform cross-sectional area floats in a liquid of density thrice its value. The portion of exposed height will be:
(1) 2/3 (2) 5/6 (3) 1/6 (4) 1/3

PPF221

14. A cubical box of wood of side 30 cm and mass 21.6 kg floats on water with two faces horizontal. The length of immersed part in water is :-
(ρ_{wood}=0.8 g/cc)
(1) 30 cm (2) 12 cm
(3) 6 cm (4) 24 cm

PPF222

15. Figure shows a container filled with a liquid of density ρ. Four points A, B, C and D lie on the diametrically opposite points of a circle as shown. Points A and C lie on vertical line and points B and D lie on horizontal line. The incorrect statement is (p_A, p_B, p_C, p_D are absolute pressure at the respective points) :-



(1) $p_D = p_B$
(2) $p_A < p_B = p_D < p_C$
(3) $p_D = p_B = \frac{p_C - p_A}{2}$
(4) $p_D = p_B = \frac{p_C + p_A}{2}$

PPF223

16. A cubical block is floating in a liquid with one fourth of its volume immersed in the liquid. If whole of the system accelerates upward with acceleration g/4, the fraction of volume immersed in the liquid will be :-
(1) 1/4 (2) 1/2 (3) 3/4 (4) 2/3

PPF224

17. A metallic sphere of mass 3 kg in air is held by a string so as to be completely immersed in a liquid of relative density 0.8. The relative density of metallic sphere is 10. The tension in the string is:- (g = 10 m/s^2)
(1) 18.7 N (2) 42.5 N
(3) 32.7 N (4) 27.6 N

PPF225

18. Water rises to a height of 10 cm in a glass capillary tube. If the area of cross-section of the tube is reduced to one-fourth of its former value then find height of water rises in the tube.
(1) 20 cm (2) 5 cm
(3) 2.5 cm (4) 7 cm

PPF226

19. The excess pressure due to surface tension inside a spherical drop is 6 unit. If eight such drops combine then the excess pressure due to surface tension inside the larger drop is :-
(1) 3 units (2) 6 units
(3) 12 units (4) 48 units

PPF227

20. Liquid rises to a height of 2 cm in a capillary tube. The angle of contact between the solid and the liquid is zero. The tube is depressed more now so that the top of the capillary is only 1 cm above the liquid then the apparent angle of contact between the solid and the liquid is:-
(1) 0° (2) 30° (3) 60° (4) 90°

PPF228

21. A frame made of metalic wire enclosing a surface area A is covered with a soap film. If the area of the frame of metallic wire is reduced by 50%, the energy of the soap film will be changed by :-
(1) 100% (2) 75% (3) 50% (4) 25%

PPF229

22. Find the difference of air pressure (in N-m^{-2}) between the inside and outside of a soap bubble 5 mm in diameter, if the surface tension is 1.6 N-m^{-1}.

(1) 2560 (2) 3720

(3) 1208 (4) 10132

PPF230

23. The surface tension of soap solution is 0.03 N/m. The work done in blowing to form a soap bubble of surface area 40 cm^2 is :-

(1) 1.2×10^{-4} J (2) 2.4×10^{-4} J

(3) 12×10^{-4} J (4) 24×10^{-4} J

PPF231

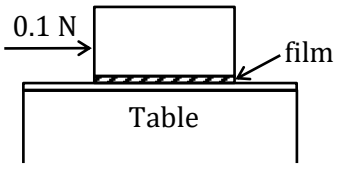
24. The height of liquid column raised in a capillary tube of certain radius when dipped in liquid A vertically is, 5 cm. If the tube is dipped in a similar manner in another liquid B of surface tension and density double the values of liquid A, the height of liquid column raised in liquid B would be _____ m.

(1) 0.20 (2) 0.5

(3) 0.05 (4) 0.10

PPF232

25. A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is 5.0×10^{-3} Pl, the speed of block is ___$\times 10^{-3}$ m/s.



PPF233

26. A mercury drop of radius 10^{-3} m is broken into 125 equal size droplets. Surface tension of mercury is 0.45 Nm^{-1}. The gain in surface energy is:

(1) 2.26×10^{-5} J (2) 28×10^{-5} J

(3) 17.5×10^{-5} J (4) 5×10^{-5} J

PPF234

27. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R) : Because the angle of contact between the oil/ grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is true but (R) is true

PPF235

ANSWER KEY

1.	(4)	**2.**	(2)	**3.**	(2)	**4.**	(1)	**5.**	(3)
6.	(3)	**7.**	(2)	**8.**	(3)	**9.**	(2)	**10.**	(4)
11.	(3)	**12.**	(1)	**13.**	(1)	**14.**	(4)	**15.**	(3)
16.	(1)	**17.**	(4)	**18.**	(1)	**19.**	(1)	**20.**	(3)
21.	(3)	**22.**	(1)	**23.**	(2)	**24.**	(3)	**25.**	25
26.	(1)	**27.**	(1)						

Properties of Matter and Fluid Mechanics-2

1. Wires A and B are made from the same material. A has twice the diameter and three times the length of B. If the elastic limits are not reached, when each is stretched by the same tension, the ratio of energy stored in A to that in B is :-

(1) 2 : 3 (2) 3 : 4 (3) 3 : 2 (4) 6 : 1

PPF236

2. The breaking stress for a wire of radius r of given material is F N/m^2. The breaking stress for the wire of same material of radius 2r is :-

(1) F/4 (2) F/2 (3) F (4) 2F

PPF237

3. The bulk modulus of rubber is 9.1 × 10^8 N/m^2. To what depth a rubber ball be taken in a lake so that its volume is decreased by 0.1%?
(g = 10 m/s^2)

(1) 25m (2) 91m
(3) 200 m (4) 500 m

PPF238

4. A solid sphere of radius R made of material of bulk modulus B is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass M is placed on the piston to compress the liquid, the fractional change in the radius of the sphere is :-

(1) $\frac{Mg}{BA}$ (2) $\frac{Mg}{3BA}$ (3) $\frac{3Mg}{4BA}$ (4) $\frac{Mg}{4BA}$

PPF239

5. If the pressure of fluid is increased from 1.01 × 10^{5}Pa to 1.165 × 10^5 Pa and volume is decreased by 10% at constant temperature, then the bulk modulus of the fluid is :-

(1) 15.5 × 10^5 Pa (2) 1.4 × 10^5 Pa
(3) 1.55 × 10^5 Pa (4) 0.0155 × 10^5 Pa

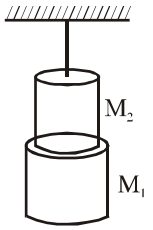
PPF240

6. The rubber cord of a catapult has cross-section area 1 mm^2 and a total unstretched length 10 cm. It is stretched to 12 cm and then released to project a particle of mass 5 g. Calculate the velocity of projection (Y for rubber is 5 × 10^8 N/m^2).

(1) 5 ms^{-1} (2) 10 ms^{-1}
(3) 20 ms^{-1} (4) 40 ms^{-1}

PPF241

7. The length of wire, when mass M_1 is hung from it, is l_1 and is l_2 with both masses M_1 and M_2 hanging. The natural length of wire is :-



(1) $\frac{M_1}{M_2}(l_1 - l_2) + l_1$ (2) $\frac{M_2 l_1 - M_1 l_2}{M_1 + M_2}$

(3) $\frac{l_1 + l_2}{2}$ (4) $\sqrt{l_1 l_2}$

PPF242

8. The Young's modulus of a wire of length L and radius r is Y newton per square metre. If the length is reduced to L/2 and radius to r/2, its Young's modulus will be :-

(1) $\frac{Y}{2}$ (2) Y (3) 2 Y (4) 4Y

PPF243

9. When a mass of 5kg is hung on a wire then extension of 30 cm take place then work done will be :- (g = 10 m/s^2)

(1) 7.5 J (2) 15 J (3) 0.5 J (4) 1.5 J

PPF244

10. The Young's modulus of brass and steel are respectively 1 × 10^{10} N/m^2 and 2 × 10^{10} N/m^2. A brass wire and a steel wire of the same length are extended by 1 mm under the same force; the radii of brass and steel wires are R_B and R_S respectively. Then :-

(1) $R_S = \sqrt{2}R_B$ (2) $R_S = \frac{R_B}{\sqrt{2}}$

(3) $R_S = 4R_B$ (4) $R_S = \frac{R_B}{4}$

PPF245

11. If ρ is the density of the material of a wire and σ is the breaking stress, the greatest length of the wire that can hang freely without breaking is:-

(1) $\frac{2\sigma}{\rho g}$ (2) $\frac{\rho}{\sigma g}$ (3) $\frac{\rho g}{2\sigma}$ (4) $\frac{\sigma}{\rho g}$

PPF246

12. The young's modulus of the material of a wire is 2×10^{10} Nm^{-2}. If the longitudinal strain is 1% then the energy stored in the wire per unit volume is :-

(1) 0.5×10^6 Jm^{-3} (2) 10^6 Jm^{-3}

(3) 2×10^6 Jm^{-3} (4) 4×10^6 Jm^{-3}

PPF247

13. A cube of side 40 cm has its upper face displaced by 0.1 mm by a tangential force of 8 kN. The shearing modulus of cube is :-

(1) 2×10^9 N-m^{-2} (2) 4×10^9 N-m^{-2}

(3) 2×10^8 Nm^{-2} (4) 4×10^8 N-m^{-2}

PPF248

14. The Bulk Modulus for an incompressible liquid is :-

(1) zero (2) unity

(3) infinity (4) between 0 and 1

PPF249

15. A steel wire of cross-sectional area 3×10^{-6} m^2 can withstand a maximum strain of 10^{-3}. Young's modulus of steel is 2×10^{11} N/m^2. The maximum mass the wire can hold is:- (Take g = 10 m/s^2)

(1) 40 kg (2) 60 kg

(3) 80 kg (4) 100 kg

PPF250

16. The Young's modulus of the material of a wire is equal to the :-

(1) stress required to increase its length four times

(2) stress required to produce unit strain

(3) strain produced in it

(4) half the strain produced in it

PPF251

17. The bulk modulus of a liquid is 3×10^{10} Nm^{-2}. The pressure required to reduce the volume of liquid by 2% is :

(1) 3×10^8 Nm^{-2} (2) 9×10^8 Nm^{-2}

(3) 6×10^8 Nm^{-2} (4) 12×10^8 Nm^{-2}

PPF252

18. The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5}m on another planet. The acceleration due to gravity on the planet will be ..... ms^{-2}. (Take acceleration due to gravity on the surface of earth = 10 m/s^{-2})

PPF253

19. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F, its length increases by 5 cm. Another wire of the same material of length 4L and radius 4r is pulled by a force 4F under same conditions. The increase in length of this wire is _____ cm.

PPF254

20. A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1kg and 2 kg are suspended respectively from its free end. Then the value of L is equal to :

(1) $\sqrt{L_1 L_2}$ (2) $\frac{L_1 + L_2}{2}$

(3) $2L_1 - L_2$ (4) $3L_1 - 2L_2$

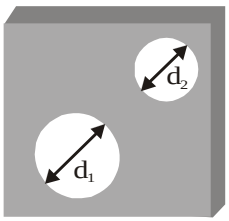
PPF255

ANSWER KEY

1.	(2)	**2.**	(3)	**3.**	(2)	**4.**	(2)	**5.**	(3)
6.	(3)	**7.**	(1)	**8.**	(2)	**9.**	(1)	**10.**	(2)
11.	(4)	**12.**	(2)	**13.**	(3)	**14.**	(3)	**15.**	(2)
16.	(2)	**17.**	(3)	**18.**	6	**19.**	5	**20.**	(3)

Thermal Physics

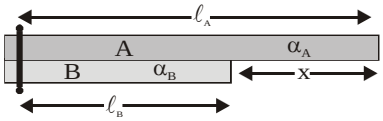
1. Two holes of unequal diameters d_1 and d_2 ($d_1 > d_2$) are cut in a metal sheet. If the sheet is heated



(1) both d_1 and d_2 will decrease
(2) both d_1 and d_2 will increase
(3) d_1 will increase, d_2 will decrease
(4) d_1 will decrease, d_2 will increase

PTM447

2. Bars of two different metals are bolted together, as shown in figure. The distance x does not change with temperature if



(1) $\frac{\ell_A}{\ell_B}=\frac{\alpha_A}{\alpha_B}$ (2) $\frac{\ell_A}{\ell_B}=\frac{\alpha_B}{\alpha_A}$

(3) $\frac{\ell_A^2}{\ell_B^2}=\frac{\alpha_A}{\alpha_B}$ (4) $\frac{\ell_A^2}{\ell_B^2}=\frac{\alpha_B}{\alpha_A}$

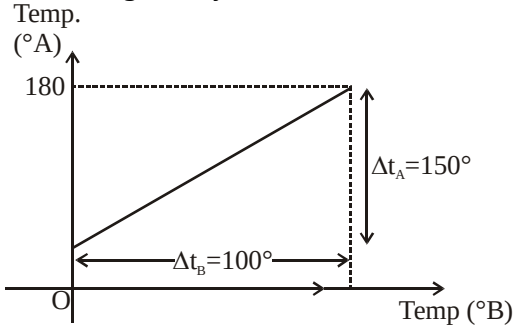
PTM448

3. Two thermometers 'X' & 'Y' shows boiling point & freezing point of water as 220°X & 20°X and 120°Y & –40°Y respectively. If 'X' shows 100°X, then find the reading in 'Y' thermometer.

(1) 25°Y (2) 50°Y (3) 20°Y (4) 24°Y

PTM449

4. The graph between two temperature scales A & B is shown between upper fixed point and lower fixed point. There are 150 divisions on scale A and 100 divisions on scale B. Then, relationship for conversion between two scales is given by:-

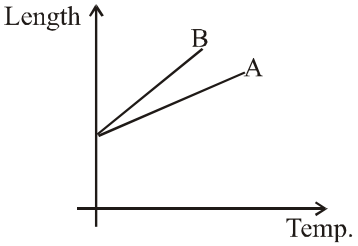


(1) $\frac{t_A-180}{100}=\frac{t_B}{150}$ (2) $\frac{t_A-30}{150}=\frac{t_B}{100}$

(3) $\frac{t_A-180}{150}=\frac{t_A}{100}$ (4) $\frac{t_B-40}{100}=\frac{t_A}{180}$

PTM450

5. Length v/s Temperature graph of A & B is given below. Find the relation between α_A & α_B.



(1) $\alpha_A > \alpha_B$ (2) $\alpha_A = \alpha_B$

(3) $\alpha_A < \alpha_B$ (4) can't say

PTM451

6. The apparent coefficient of expansion of a liquid when heated in a copper vessel is C and when heated in a silver vessel is S. If A is the coefficient of linear expansion of copper, the coefficient of linear expansion of silver is :-

(1) $\frac{C+S-3A}{3}$ (2) $\frac{C+3A-S}{3}$

(3) $\frac{S+3A-C}{3}$ (4) $\frac{C+S+3A}{3}$

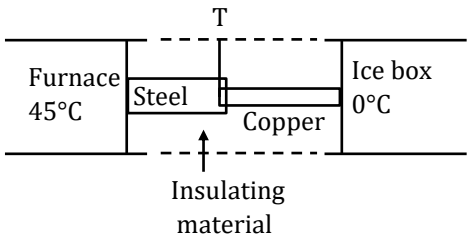
PTM452

7. A unit scale is to be prepared whose length does not change with temperature and remains 20 cm, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is 40 cm and length of iron will be _____cm.
($\alpha_{iron} = 1.2 \times 10^{-5}\ K^{-1}$ and $\alpha_{brass} = 1.8 \times 10^{-5}\ K^{-1}$).

PTM453

8. If K_1 and K_2 are the thermal conductivities L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1}=9, \frac{A_1}{A_2}=2, \frac{L_1}{L_2}=2$.

Then, for the arrangement as shown in the figure. The value of temperature T of the steel – copper junction in the steady state will be :



(1) 18 °C (2) 14 °C
(3) 4.5 °C (4) 150 °C

PTM454

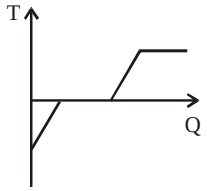
9. A faulty thermometer reads 5°C in melting ice and 95°C in steam. The correct temperature on absolute scale will be..... K when the faulty thermometer reads 41°C.

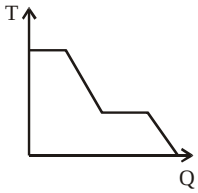
PTM455

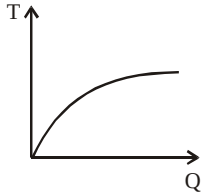
10. A thin rod having a length of 1 m and area of cross-section $3\times10^{-6}\,m^2$ is suspended vertically from one end. The rod is cooled from 210°C to 160°C. After cooling, a mass M is attached at the lower end of the rod such that the length of rod again becomes 1 m. Young's modulus and coefficient of linear expansion of the rod are $2\times10^{11}\,N\,m^{-2}$ and $2\times10^{-5}K^{-1}$, respectively. The value of M is_____ kg. (Take g = 10 m s^{-2})

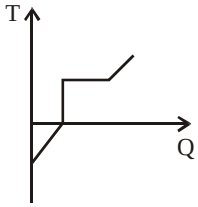
PTM456

11. A block of ice at –12°C is slowly heated and converted into steam at 100°C. Which of the following curves best represents the event?

(1) 

(2) 

(3) 

(4) 

PTM457

12. If 10 g of ice at 0°C is mixed with 10 g of water at 40°C. The final mass of water in mixture is (Latent heat of fusion of ice = 80 cal/g; specific heat of water = 1 cal/g °C)

(1) 10 g (2) 15 g
(3) 18 g (4) 20 g

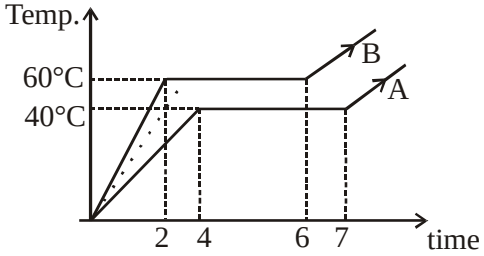
PTM458

13. A 10 g ice cube is dropped into 45 g of water kept in a glass. If the water was initially at a temperature of 28°C and the temperature of ice –15°C, find the final temperature (in °C) of water.
(Specific heat of ice = 0.5 cal/g –°C and L = 80 cal/g)

(1) 14 (2) 7 (3) 28 (4) None

PTM459

14. In figure given two solid A and B of same mass are heated at constant rate by same source then find



(i) ratio of latent heat for fusion $\left(\frac{L_A}{L_B}\right)$

(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{3}{5}$

(ii) ratio of specific heat for solid state $\left(\frac{S_A}{S_B}\right)$

(1) $\frac{1}{3}$ (2) 3 (3) 6 (4) $\frac{1}{6}$

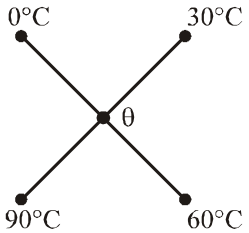
PTM460

15. Ice at –20°C mixed with 200g water at 25°C. If temperature of mixture is 10°C then mass of ice is –

(1) 30 gm (2) 20 gm (3) 15 gm (4) 40 gm

PTM461

16. Four rods of same material and having the same cross section and length have been joined, as shown. The temperature of the junction of four rods will be :



(1) 20°C (2) 30°C (3) 45°C (4) 60°C

PTM462

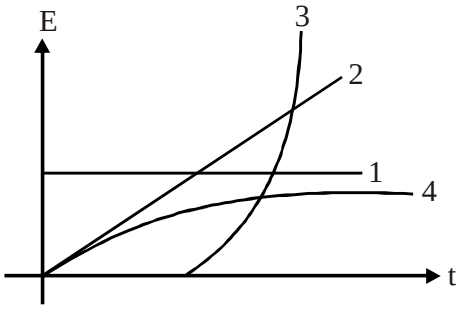
17. Two electric lamps A and B radiate the same power. Their filaments have the same dimensions, and have emissivities e_A and e_B. Their surface temperatures are T_A and T_B. The ratio $\frac{T_A}{T_B}$ will be equal to

(1) $\left(\frac{e_B}{e_A}\right)^{1/4}$ (2) $\left(\frac{e_B}{e_A}\right)^{1/2}$

(3) $\left(\frac{e_A}{e_B}\right)^{1/2}$ (4) $\left(\frac{e_A}{e_B}\right)^{1/4}$

PTM463

18. A ball of surface temperature T is in thermal equilibrium with its environment. Which of the curve gives the radiating power E radiated by the sphere as a function of time 't'?



(1) 1 (2) 2

(3) 3 (4) 4

PTM464

19. Two stars emit maximum radiation at wavelength 3600 Å and 4800 Å respectively. The ratio of their temperatures is –

(1) 1 : 2 (2) 3 : 4

(3) 4 : 3 (4) 2 : 1

PTM465

20. A heated body maintained at T K emits thermal radiation of total energy E with a maximum intensity at frequency ν. The emissivity of the material is 0.5. If the temperature of the body be increased and maintained at temperature 3T K, then :-

(i) The maximum intensity of the emitted radiation will occur at frequency ν/3

(ii) The maximum intensity of the emitted radiation will occur at frequency 3ν.

(iii) The total energy of emitted radiation will become 81 E

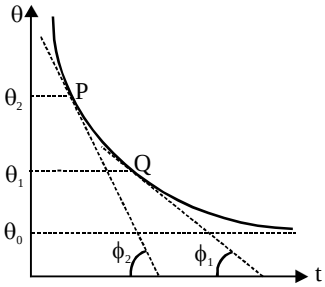
(iv) The total energy of emitted radiation will become 27 E

(1) (i) & (ii) (2) (ii) & (iii)

(3) (i) & (iv) (4) (i), (ii) & (iv)

PTM466

21. A body cools in a surrounding which is at a constant temperature of θ_0. Assume that it obeys Newton's law of cooling. Its temperature θ is plotted against time t. Tangents are drawn to the curve at the points $P(\theta = \theta_2)$ and $Q(\theta = \theta_1)$. These tangents meet the time axis at angles of ϕ_2 and ϕ_1, as shown

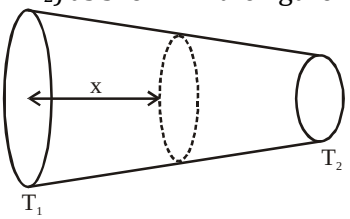


(1) $\frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_1 - \theta_0}{\theta_2 - \theta_0}$ (2) $\frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$

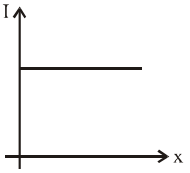
(3) $\frac{\tan\phi_1}{\tan\phi_2} = \frac{\theta_1}{\theta_2}$ (4) $\frac{\tan\phi_1}{\tan\phi_2} = \frac{\theta_2}{\theta_1}$

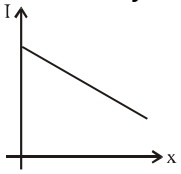
PTM467

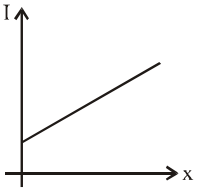
22. Two ends of a rod of non uniform area of cross-section are maintained at temperature T_1 and T_2 ($T_1 > T_2$) as shown in the figure.

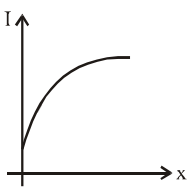


If I is heat current through the cross-section of conductor at distance x from its left face, then the variation of I with x is best represented by:

(1) 

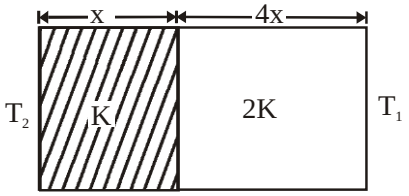
(2) 

(3) 

(4) 

PTM468

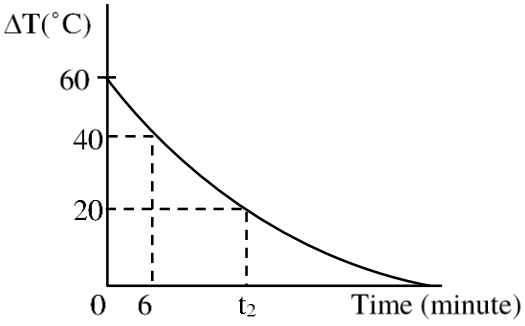
23. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left[\frac{A(T_2 - T_1)K}{x}\right]$ f, with f equal to: -



(1) 1 (2) $\frac{1}{2}$ (3) $\frac{2}{3}$ (4) $\frac{1}{3}$

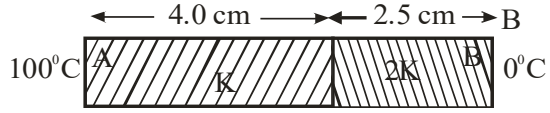
PTM469

24. In an experiment to verify Newton's law of cooling, a graph is plotted between, the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C. The value of t_2 as mentioned in the graph will be _____.



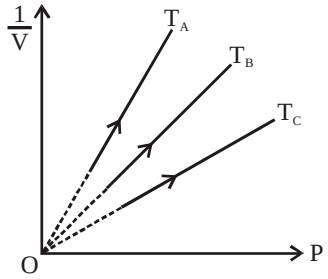
PTM470

25. As per the given figure, two plates A and B of thermal conductivity K and 2 K are joined together to form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1+\frac{5}{\alpha}\right)$K, then the value of α will be _____.



PTM471

26. Figure shows the isotherms of fixed mass of an ideal gas at three temperatures T_A, T_B and T_C then.



(1) $T_A > T_B > T_C$ (2) $T_A < T_B < T_C$
(3) $T_B < T_A < T_C$ (4) $T_A = T_B = T_C$

PTM472

27. The pressure (P) and absolute temperature (T) of an ideal gas are related to each other as $P \propto \frac{1}{T^2}$. If temperature of gas is 300 K and its volume changes from V to 8 V, then find the final temperature of gas.

(1) 600 K (2) 300 K
(3) 373 K (4) 273 K

PTM473

28. Four molecules of a gas have speeds 1, 2, 3 and 4 km/s. The value of the r.m.s. speed of the gas molecule is :-

(1) $\frac{1}{2}\sqrt{15}$ km/s (2) $\frac{1}{2}\sqrt{10}$ km/s
(3) 2.5 km/s (4) $\sqrt{\frac{15}{2}}$ km/s

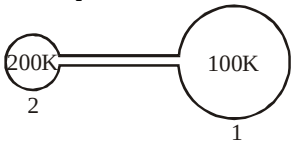
PTM474

29. The rms speed of helium gas at 27°C and 1 atm pressure is 900 ms^{-1}. Then the rms speed of helium molecules at temperature 27°C and 2 atm pressure is

(1) 450 m/s (2) 900 m/s
(3) 1800 m/s (4) 750 m/s

PTM475

30. Figure shows two flasks connected to each other. The volume of the flask 1 is twice that of flask 2. The system is filled with an ideal gas at temperature 100 K and 200 K respectively in the flasks. If the mass of the gas in 1 be m, then what is the mass of the gas in flask 2 in equilibrium?



(1) m (2) $\frac{m}{2}$ (3) $\frac{m}{4}$ (4) $\frac{m}{8}$

PTM476

31. One mole of a diatomic gas undergoes a process $P=\frac{P_0}{1+\left(\frac{V}{V_0}\right)^3}$, where P_0, V_0 are constants. The translational kinetic energy of the gas when $V = V_0$ is given by

(1) $\frac{5P_0V_0}{4}$ (2) $\frac{3P_0V_0}{4}$

(3) $\frac{3P_0V_0}{2}$ (4) $\frac{5P_0V_0}{2}$

PTM477

32. An ideal gas has an initial pressure of 3 pressure units and an initial volume of 4 volume units. The table gives the final pressure and volume of the gas (in those same units) in four processes. Which state lies on the same isotherm

	A	B	C	D
P	5	4	12	6
V	7	6	1	3

(1) A (2) B (3) C (4) D

PTM478

33. **Statement-1 :** At low pressure and high temperature real gas approaches the ideal gas behaviour.

and

Statement-2 : At low pressure and high temperature the molecules are far apart and molecular interactions are negligible.

(1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1

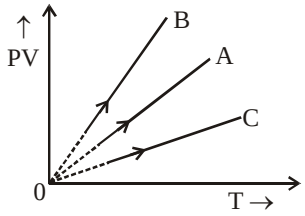
(2) Statement-1 is true, Statement-2 is true, Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

PTM479

34. PV versus T graph of equal masses of H_2, He and O_2 is shown in figure. Choose the correct alternative.



(1) A corresponds to He, B to H_2 and C to O_2

(2) A corresponds to H_2, B to He and C to O_2

(3) A corresponds to He, B to O_2 and C to H_2

(4) A corresponds to O_2, B to H_2 and C to He

PTM480

35. At 20°C temperature, an argon gas at atmospheric pressure is confined in a vessel with a volume of $1m^3$, The effective hard-sphere diameter of argon atom is 3.10×10^{-10}m. It's mean free path is :

(1) 100 nm (2) 90 nm

(3) 93.6 nm (4) 95 nm

PTM481

36. Same gas is filled in two vessels of the same volume at the same temperature. If the ratio of the number of molecules is 1:4, then

A. The r.m.s. velocity of gas molecules in two vessels will be the same.

B. The ratio of pressure in these vessels will be 1 : 4

C. The ratio of pressure will be 1 : 1

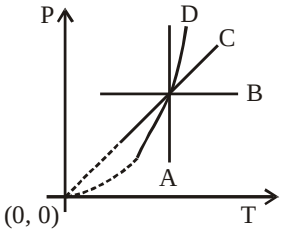
D. The r.m.s. velocity of gas molecules in two vessels will be in the ratio of 1 : 4

(1) A and C only (2) B and D only

(3) A and B only (4) C and D only

PTM482

37. Following figure shows P-T graph for four processes A, B, C and D. Select the correct alternative



(1) A-Isobaric process

(2) B-Adiabatic process

(3) C-Isochoric process

(4) D-Isothermal process

PTM483

38. A flask contains argon and oxygen in the ratio of 3:2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively will be :

(1) 3 : 2 (2) 9 : 4

(3) 2 : 3 (4) 3 : 5

PTM484

39. According to kinetic theory of gases,

A. The motion of the gas molecules freezes at 0°C

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases)

Choose the most appropriate answer from the options given below:

(1) A and C only (2) B and C only

(3) A and B only (4) C and D only

PTM485

40. What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(1) The velocity of atomic oxygen remains same

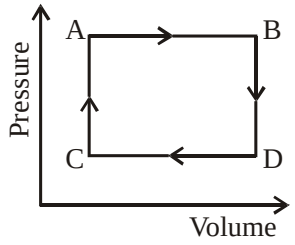
(2) The velocity of atomic oxygen doubles

(3) The velocity of atomic oxygen becomes half

(4) The velocity of atomic oxygen becomes four times

PTM486

41. The pressure and volume of gas are changed as shown in the P-V diagram in the figure ahead. The temperature of the gas :



(1) Increases as it goes from A to B

(2) Increases as it goes from B to C

(3) Remains constant during these changes

(4) Decreases as it goes from D to A

PTM487

42. An ideal refrigerator runs between –23°C and 27°C. Find the heat rejected to atmosphere for every joule of work input.

(1) 6 J (2) 5 J (3) 4 J (4) 3 J

PTM488

43. Select the incorrect relation. (Where symbols have their usual meanings) :-

(1) $C_P = \frac{\gamma R}{\gamma - 1}$ (2) $C_P - C_V = R$

(3) $\Delta U = \frac{P_f V_f - P_i V_i}{1-\gamma}$ (4) $C_V = \frac{R}{\gamma - 1}$

PTM489

44. Select the incorrect statement about the specific heat of a gaseous system :

(1) Specific heat at no heat exchange condition, $C_A = 0$

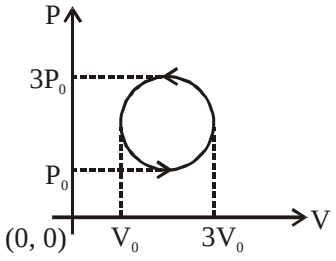
(2) Specific heat at constant temperature, $C_T = \infty$

(3) Specific heat at constant pressure, $C_P = \frac{\gamma R}{\gamma - 1}$

(4) Specific heat at constant volume, $C_v = \frac{R}{\gamma}$

PTM490

45. Work done in the cyclic process shown in figure is :-



(1) $4P_0V_0$ (2) $-4P_0V_0$

(3) $-\frac{22}{7}P_0V_0$ (4) $-13P_0V_0$

PTM491

46. The specific heat of a gas in a polytropic process (PV^x = constant) is given by :

(1) $C_V + \frac{R}{1+x}$ (2) $C_P + \frac{R}{1+x}$

(3) $C_V + \frac{R}{1-x}$ (4) $C_P + \frac{R}{1-x}$

PTM492

47. One mole of hydrogen, assumed to be ideal, is adiabatically expanded from its initial state (P_1, V_1, T_1) to the final state (P_2,V_2,T_2). The decrease in the internal energy of the gas during this process will be given by

(1) $C_v (T_1 - T_2)$

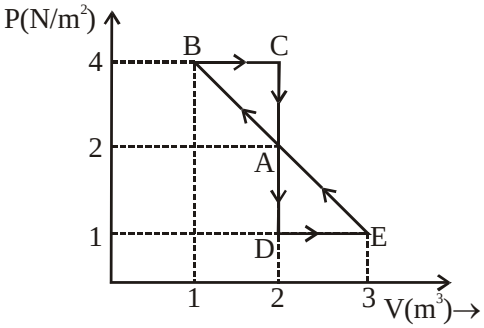
(2) $C_p(T_1 - T_2)$

(3) $\frac{C_P + C_V}{2}(T_1 - T_2)$

(4) $(C_p - C_v) (T_1 - T_2)$

PTM493

48. One mole of a monoatomic gas is carried along process ABCDEA as shown in the diagram. Find the net work done by gas :-



(1) $\frac{3}{2}$J (2) 1J (3) $\frac{1}{2}$J (4) 0J

PTM494

49. A Carnot engine whose heat sinks at 27°C, has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?

(1) Increases by 18°C

(2) Increase by 200°C

(3) Increase by 120°C

(4) Increase by 73°

PTM495

50. The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by:

(1) $\left(1+\frac{f}{3}\right)$ (2) $\left(1+\frac{2}{f}\right)$

(3) $\left(1+\frac{f}{2}\right)$ (4) $\left(1+\frac{1}{f}\right)$

PTM496

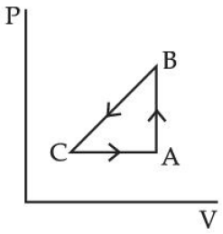
51. A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel temperature of the gas increases by :
(R = universal gas constant)

(1) $\frac{Mv^2}{7R}$ (2) $\frac{Mv^2}{5R}$

(3) $2\frac{Mv^2}{7R}$ (4) $7\frac{Mv^2}{5R}$

PTM497

52. A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60J of heat during CA. A work 50J is done on the gas during the part BC. The internal energy of the gas at A is 1560J. The work done by the gas during the part CA is:



(1) 20 J (2) 30 J (3) –30J (4) –60 J

PTM498

53. Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal. W_2. if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

(1) $W_1 < W_2 < W_3$ (2) $W_2 < W_3 < W_1$

(3) $W_3 < W_1 < W_2$ (4) $W_2 < W_1 < W_3$

PTM499

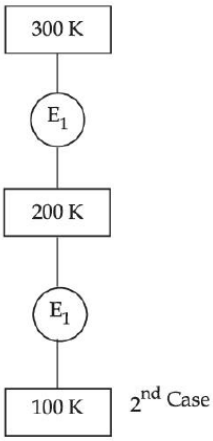
54. 7 mole of certain monoatomic ideal gas undergoes a temperature increase of 40K at constant pressure. The increase in the internal energy of the gas in this process is
(Given R = 8.3 $JK^{-1}mol^{-1}$)

(1) 5810 J (2) 3486 J

(3) 11620J (4) 6972 J

PTM500

55. In 1st case, Carnot engine operates between temperatures 300 K and 100 K. In 2nd case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2nd case) will be :



(1) same as the 1st case
(2) always greater than the 1st case
(3) always less than the 1st case
(4) may increase or decrease with respect to the 1st case

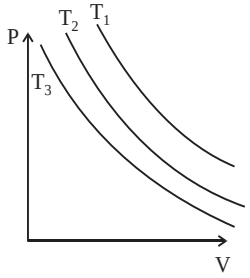
PTM501

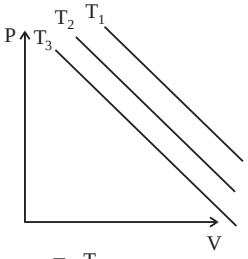
56. The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2(>P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes _____ times of its initial temperature.

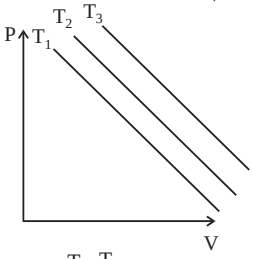
(given $\frac{d_2}{d_1} = 32$)

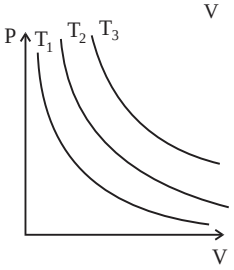
PTM502

57. In an Isothermal change, the change in pressure and volume of a gas can be represented for three different temperature; $T_3 > T_2 > T_1$ as :

(1)


(2)


(3)


(4)


PTM503

58. A Carnot engine with efficiency 50% takes heat from a source at 600 K. In order to increase the efficiency to 70%, keeping the temperature of sink same, the new temperature of the source will be :

(1) 360 K (2) 1000 K
(3) 900 K (4) 300 K

PTM504

59. Match List I with List II :

List I		List II	
A.	Isothermal Process	I.	Work done by the gas decreases internal energy
B.	Adiabatic Process	II.	No change in internal energy
C.	Isochoric Process	III.	The heat absorbed goes partly to increase internal energy and partly to do work
D.	Isobaric Process	IV.	No work is done on or by the gas

Choose the correct answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-I, B-II, C-III, D-IV

PTM505

60. Heat is given to an ideal gas in an isothermal process.

A. Internal energy of the gas will decrease.

B. Internal energy of the gas will increase.

C. Internal energy of the gas will not change.

D. The gas will do positive work.

E. The gas will do negative work.

Choose the **correct** answer from the options given below :

(1) A and E only

(2) B and D only

(3) C and E only

(4) C and D only

PTM506

ANSWER KEY

1.	(2)	**2.**	(2)	**3.**	(4)	**4.**	(2)	**5.**	(3)
6.	(2)	**7.**	60	**8.**	(3)	**9.**	313	**10.**	60
11.	(1)	**12.**	(2)	**13.**	(2)	**14.**	(i) – 1, (ii) – 2	**15.**	(1)
16.	(3)	**17.**	(1)	**18.**	(1)	**19.**	(3)	**20.**	(2)
21.	(2)	**22.**	(1)	**23.**	(4)	**24.**	16	**25.**	21
26.	(2)	**27.**	(1)	**28.**	(4)	**29.**	(2)	**30.**	(3)
31.	(1)	**32.**	(3)	**33.**	(1)	**34.**	(1)	**35.**	(3)
36.	(3)	**37.**	(3)	**38.**	(4)	**39.**	(2)	**40.**	(2)
41.	(1)	**42.**	(1)	**43.**	(3)	**44.**	(4)	**45.**	(3)
46.	(3)	**47.**	(1)	**48.**	(3)	**49.**	(2)	**50.**	(2)
51.	(2)	**52.**	(2)	**53.**	(4)	**54.**	(2)	**55.**	(1)
56.	(4)	**57.**	(4)	**58.**	(2)	**59.**	(2)	**60.**	(4)

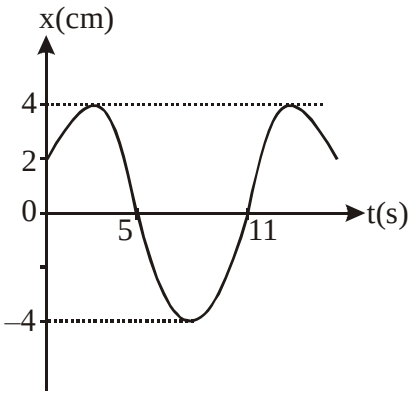
Oscillations

1. A particle is executing SHM with time period T Starting from mean position, time taken by it to complete $\frac{5}{8}$ oscillations is,

(1) $\frac{T}{12}$ (2) $\frac{T}{6}$ (3) $\frac{5T}{12}$ (4) $\frac{7T}{12}$

POS200

2. Figure shows the position-time graph of an object in SHM. The correct equation representing motion is :-



(1) $2\sin\left(\frac{2\pi}{5}t+\frac{\pi}{6}\right)$ (2) $4\sin\left(\frac{\pi}{5}t+\frac{\pi}{6}\right)$

(3) $4\sin\left(\frac{\pi}{6}t+\frac{\pi}{3}\right)$ (4) $4\sin\left(\frac{\pi}{6}t+\frac{\pi}{6}\right)$

POS201

3. The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

(1) 6 s (2) 8 s (3) 12s (4) 36 s

POS202

4. The length of a seconds pendulum at a height h = 2R from earth surface will be :
(Given : R = Radius of earth and acceleration due to gravity at the surface of earth $g = \pi^2 ms^{-2}$)

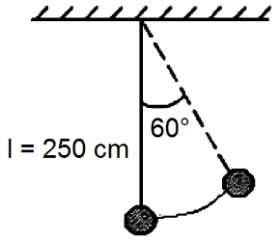
(1) $\frac{2}{9}m$ (2) $\frac{4}{9}m$ (3) $\frac{8}{9}m$ (4) $\frac{1}{9}m$

POS203

5. A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6s. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is _____ s.

POS204

6. A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200 g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob. the maximum velocity attained by the bob will be _____ms^{-1}. (if $g = 10\ m/s^2$)



POS205

7. A body makes angular SHM of amplitude $\frac{\pi}{10}$ rad. and time period 0.05 s. If the body is at displacement $\theta = \frac{\pi}{10}$ rad. at t = 0, then write the equation giving the angular displacement as a function of time

POS206

8. The acceleration of a certain simple harmonic oscillator is given by

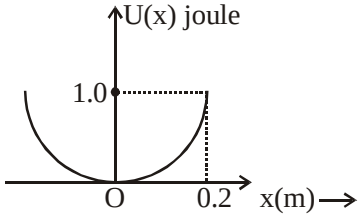
$$a = -(35.28\ m/s^2)\cos 4.2t$$

The amplitude of the simple harmonic motion is

(1) 2.0 m (2) 8.4 m
(3) 16.8 m (4) 17.64 m

POS207

9. A particle of mass 4kg moves simple harmonically such that its PE (U) varies with position x, as shown. The period of oscillations is :-



(1) $\frac{2\pi}{25}s$ (2) $\frac{\pi\sqrt{2}}{5}s$

(3) $\frac{4\pi}{5}s$ (4) $\frac{2\pi\sqrt{2}}{5}s$

POS208

10. F-x and x – t graph of a particle in SHM are as shown in figure match the following

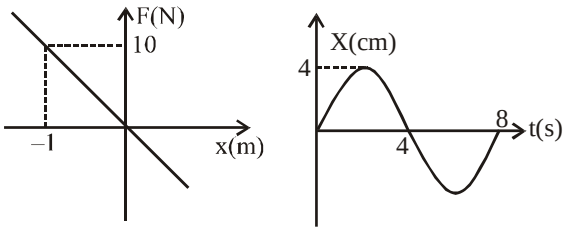


Table-1		Table-2	
(A)	Mass of the particle	(p)	$\frac{\pi}{2}$ SI unit
(B)	Maximum kinetic energy of particle	(q)	$\left(\frac{160}{\pi^2}\right)$ SI units
(C)	Angular frequency of particle	(r)	(8.0×10^{-3}) SI unit
		(s)	None

POS209

11. A particle is moving on x–axis has potential energy U = 2 – 20x + 5 x^2 joules. The particle is released at x = – 3. The maximum value of 'x' will be [x is in meters and U is in joules]:-

(1) 5 m (2) 3 m

(3) 7 m (4) 8 m

POS210

12. A particle of mass m performs SHM along a straight line with frequency f and amplitude A: -

(1) The average kinetic energy of the particle is zero

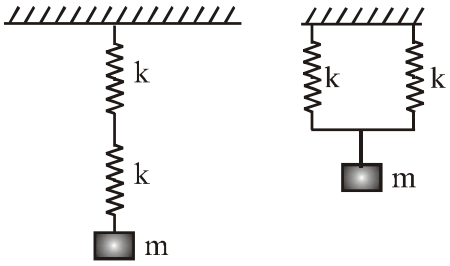
(2) The average potential energy is $m\pi^2f^2A^2$

(3) The frequency of oscillation of kinetic energy is 2f

(4) Velocity function leads acceleration in phase by $\pi/2$

POS211

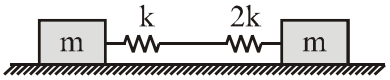
13. Two identical springs of constant k are connected in series and parallel as shown in figure. A mass' m' is suspended from them. The ratio of their frequencies of vertical oscillations will be :-



(1) 2 : 1 (2) 1 : 1 (3) 1 : 2 (4) 4 : 1

POS212

14. A system is shown in the figure. The time period for small oscillations of the two blocks will be :-



(1) $2\pi\sqrt{\frac{3m}{k}}$ (2) $2\pi\sqrt{\frac{3m}{2k}}$

(3) $2\pi\sqrt{\frac{3m}{4k}}$ (4) $2\pi\sqrt{\frac{3m}{8k}}$

POS213

15. A mass M is suspended with a light spring. An additional mass m added displaces the spring further by a distance x. Now, the combined mass will oscillate on the spring with period :

(1) $T=2\pi\sqrt{\frac{mg}{x(M+m)}}$ (2) $T=2\pi\sqrt{\frac{(M+m)x}{mg}}$

(3) $T=\frac{\pi}{2}\sqrt{\frac{m}{x(M+m)}}$ (4) $T=2\pi\sqrt{\frac{M+m}{mgx}}$

POS214

16. As per given figures, two springs of spring constants K and 2K are connected to mass m. If the period of oscillation in figure (a) is 3s, then the period of oscillation in figure (b) will be $\sqrt{x}$ s. The value of x is_____.

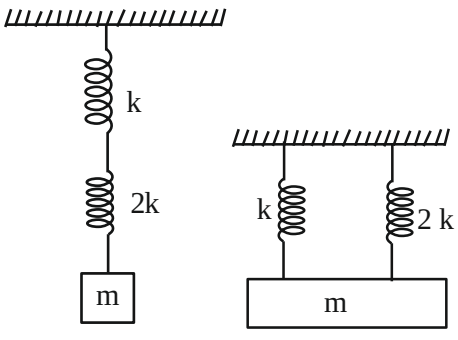


figure (a) figure (b)

POS215

17. A force of 6.4N stretches a vertical spring by 0.1m. The mass that must be suspended from the spring so that it oscillates with a time period of $\pi/4$ second :-

(1) $\frac{\pi}{4}$kg (2) $\frac{4}{\pi}$kg (3) 1 kg (4) 10 kg

POS216

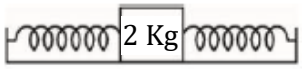
18. The potential energy of a particle of mass 4 kg in motion along the x-axis is given by U = 4(1 – cos 4x) J. The time period of the particle for small oscillation ($\sin\theta \simeq \theta$) is $\left(\frac{\pi}{K}\right)$s. The value of K is .....

POS217

19. The amplitude of a particle executing SHM is 3 cm. The displacement at which its kinetic energy will be 25% more than the potential energy is: _____ cm.

POS218

20. A block of mass 2 kg is attached with two identical springs of spring constant 20 N/m each. The block is placed on a frictionless surface and the ends of the springs are attached to rigid supports (see figure). When the mass is displaced from its equilibrium position, it executes a simple harmonic motion. The time period of oscillation is $\frac{\pi}{\sqrt{x}}$ in SI unit. The value of x is _____ .



POS219

21. The general displacement of a simple harmonic oscillator is $x = A\sin\omega t$. Let T be its time period. The slope of its potential energy (U) – time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is ___ .

POS220

22. The maximum potential energy of a block executing simple harmonic motion is 25 J. A is amplitude of oscillation. At $A/2$, the kinetic energy of the block is

(1) 37.5 J (2) 9.75 J
(3) 18.75 J (4) 12.5 J

POS221

23. The time period of oscillations of a simple pendulum is 1 minute. If its length is increased by 44%, then its new time period of oscillations will be :-

(1) 96s (2) 58s (3) 82s (4) 72s

POS222

24. A simple pendulum of length 40 cm oscillates with an angular amplitude of 0.04 rad. Find

(a) The time period
(b) The linear amplitude of the bob
(c) The speed of the bob when the string makes 0.02 rad with the vertical
(d) The angular acceleration when the bob is in momentary rest (take g = 10 m/s^2)

POS223

25. The time period and the amplitude of a simple pendulum are 4 seconds and 0.20 meter respectively. If the displacement is 0.1 meter at time t = 0, the equation of its displacement is represented by :-

(1) y = 0.2 sin (0.5πt)

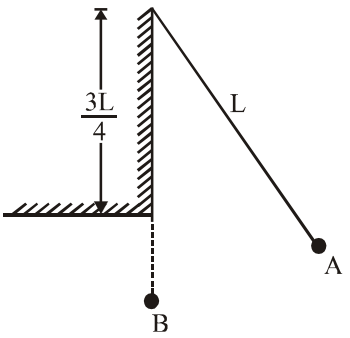
(2) y = 0.1 sin $(0.5\pi t + \frac{\pi}{6})$

(3) y = 0.1 sin $\left(\pi t + \frac{\pi}{6}\right)$

(4) y = 0.2 sin $\left(0.5\pi t + \frac{\pi}{6}\right)$

POS224

26. A pendulum has period T for small oscillations. An obstacle is placed directly beneath the pivot, so that only the lowest one quarter of the string can follow the pendulum bob when it swings in the left of its resting position as shown in the figure. The pendulum is released from rest at a certain point A. The time taken by it to return to that point is :-



(1) T (2) T/2
(3) 3T/4 (4) T/4

POS225

27. The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, the time period of oscillation would :-

(1) remain unchanged

(2) increase towards a saturation value

(3) first increase and then decrease to the original value

(4) first decrease and then increase to the original value

POS226

28. Choose the correct statement :-

(1) Time period of a simple pendulum depends on amplitude.

(2) Time shown by a spring watch varies with acceleration due to gravity g.

(3) In a simple pendulum, time period varies linearly with the length of the pendulum.

(4) The graph between length of the pendulum and time period is a parabola.

POS227

29. A simple pendulum has a time period T in vacuum. Its time period when it is completely immersed in a liquid of density $\frac{1}{8}$th of material of the bob is :-

(1) $\sqrt{\frac{7}{8}}T$ (2) $\sqrt{\frac{5}{8}}T$

(3) $\sqrt{\frac{3}{8}}T$ (4) $\sqrt{\frac{8}{7}}T$

POS228

30. The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by : –

(A) $2\pi\sqrt{L/(g\cos\alpha)}$

(B) $2\pi\sqrt{L/(g\sin\alpha)}$

(C) $2\pi\sqrt{L/g}$

(D) $2\pi\sqrt{L/(g\tan\alpha)}$

POS229

ANSWER KEY

1.	(4)	**2.**	(4)	**3.**	(4)	**4.**	(4)	**5.**	1
6.	5	**7.**	$\left[\theta=\frac{\pi}{10}\cos 40\pi t\right]$			**8.**	(1)	**9.**	(4)
10.	[A-q, B-r, C-s]	**11.**	(3)	**12.**	(2,3)	**13.**	(3)	**14.**	(3)
15.	(2)	**16.**	2	**17.**	(3)	**18.**	2	**19.**	2
20.	5	**21.**	8	**22.**	(3)	**23.**	(4)		
24.	[1.26 s, 1.6 cm, 6.8 cm/s, 1 rad/s^2]			**25.**	(4)	**26.**	(3)	**27.**	(3)
28.	(4)	**29.**	(4)	**30.**	(A)				

Wave Motion

1. Two particles of medium disturbed by the wave propagation are at $x_1 = 0$ and $x_2 = 1$cm. The respective displacements (in cm) of the particles can be given by the equations: -

$y_1 = 2\sin 3\pi t$

$y_2 = 2\sin(3\pi t - \pi/8)$

The wave velocity is :

(1) 16 cm/sec (2) 24 cm/sec

(3) 12 cm/sec (4) 8 cm/sec

PWA230

2. Two small boats are 10m apart on a lake. Each pops up and down with a period of 4.0 seconds due to wave motion on the surface of water. When one boat is at its highest point, the other boat is at its lowest point. Both boats are always within a single cycle of the waves. The speed of the waves is :-

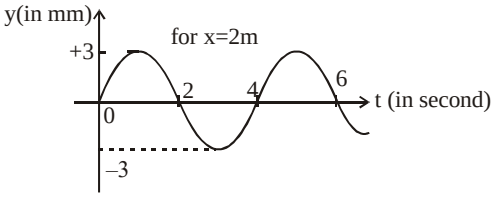
(1) 2.5 m/s (2) 5.0 m/s

(3) 14 m/s (4) 40 m/s

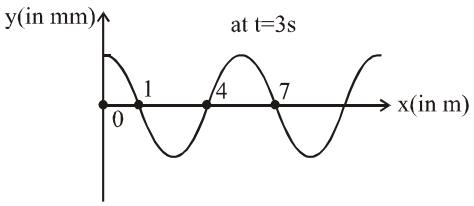
PWA231

3. A sinusoidal wave propagates along a string. In figure (a) and (b) 'y' represents displacement of particle from the mean position. 'x' & 't' have usual meanings. Find :-

(a)



(b)

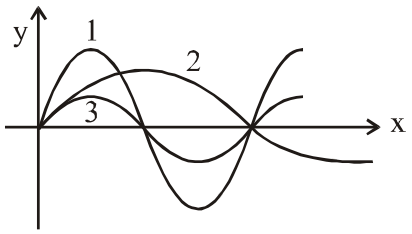


(a) wavelength, frequency and speed of the wave

(b) maximum velocity and maximum acceleration of the particles

PWA232

4. Graph shows three waves that are separately sent along a string that is stretched under a certain tension along x-axis. If ω_1, ω_2 and ω_3 are their angular frequencies respectively then :-

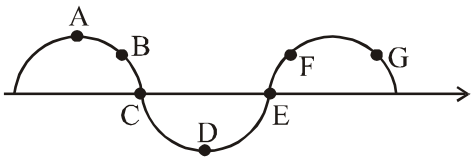


(1) $\omega_1 = \omega_3 > \omega_2$ (2) $\omega_1 > \omega_2 > \omega_3$

(3) $\omega_1 > \omega_2 = \omega_3$ (4) $\omega_1 = \omega_2 = \omega_3$

PWA233

5. The following figure depicts a wave travelling in a medium. Which pair of particles are in phase :-



(1) A and D (2) B and F

(3) C and E (4) B and G

PWA234

6. A transverse periodic wave on a string with a linear mass density of 0.200 kg/m is described by the following equation y = 0.05 sin (420t – 21.0x) where x and y are in metres and t is in seconds. The tension in the string is equal to :

(1) 32 N (2) 42 N (3) 66 N (4) 80 N

PWA235

7. The velocity of sound in a gas at temperature 27°C is V then in the same gas its velocity will be 2V at temperature :-

(1) 54°C (2) 327°C

(3) 927°C (4) 108°C

PWA236

8. Consider the wave represented by,

$y = \cos(500\,t - 70\,x)$ where y is in mm, x in m and t in seconds. Which of the following is/are true?

(1) the wave is a standing wave

(2) the speed of the wave is $50/7$ ms^{-1}

(3) the frequency of oscillation is $500 \times 2\pi$ Hz

(4) two nearest points in the same phase have separation $20\pi/7$cm

PWA237

9. Which of the following expressions is that of a simple harmonic progressive wave?

(1) A sin ωt (2) A sin ωt cos kx

(3) A sin (ωt – kx) (4) A cos kx

PWA238

10. In the wave equation

$$y = 0.5\sin\frac{2\pi}{\lambda}(400t - x)m$$

the velocity of the wave will be :

(1) 200 m/s (2) $200\sqrt{2}\,m/s$

(3) 400 m/s (4) $400\sqrt{2}\,m/s$

PWA239

11. Two coherent sources of different intensities send waves which interfere. The ratio of the maximum intensity to the minimum intensity is 25. The intensities are in the ratio :-

(1) 25 : 1 (2) 5 : 1 (3) 9 : 4 (4) 625 :1

PWA240

12. When two progressive waves of intensity I_1 and I_2 but slightly different frequencies superpose, the resultant intensity fluctuates between: -

(1) $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$ and $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$

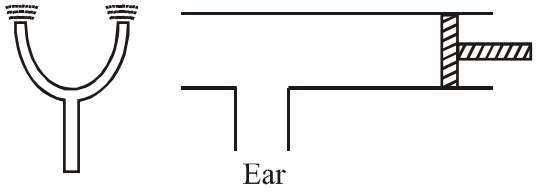
(2) $\left(\sqrt{I_1}+\sqrt{I_2}\right)$and $\left(\sqrt{I_1}-\sqrt{I_2}\right)$

(3) $(I_1 + I_2)$ and $\left(\sqrt{I_1}-\sqrt{I_2}\right)$

(4) $\frac{I_1}{I_2}$and$\frac{I_2}{I_1}$

PWA241

13. A vibrating tuning fork of frequency n is placed near the open end of a long cylindrical tube. The tube has a side opening and is also fitted with a movable reflecting piston. As the piston is moved through 8.75 cm, the intensity of sound changes from a maximum to minimum. If the speed of sound is 350 metre per second, then n is :-



(1) 500 Hz (2) 1000 Hz

(3) 2000 Hz (4) 4000 Hz

PWA242

14. The path difference between the two waves

$y_1 = a_1 \sin(\omega t - kx)$

and $y_2 = a_2 \cos(\omega t - kx + \phi)$,

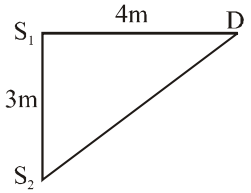
is

(1) $(\lambda/2\pi)\phi$ (2) $\left(\frac{\phi+(\pi/2)}{2\pi}\right)\lambda$

(3) $\frac{2\pi}{\lambda}\left(\phi-\frac{\pi}{2}\right)$ (4) $\frac{2\pi}{\lambda}\phi$

PWA243

15. In the figure the intensity of waves arriving at D from two coherent sources S_1 and S_2 is I_0. The wavelength of the wave is λ = 4m. Resultant intensity at D will be :-



(1) $4I_0$ (2) I_0 (3) $2I_0$ (4) zero

PWA244

16. When a guitar is sounded with a 440 Hz tuning fork, a beat frequency of 5 Hz is heard. If the experiment is repeated with a tuning fork of 437 Hz, the beat frequency is 8 Hz. The string frequency (in Hz) is :-

(1) 445 (2) 435

(3) 429 (4) 448

PWA245

17. The equations of two waves are given by :

$$y_1 = 5\sin 2\pi(x - vt)cm$$

$$y_2 = 3\sin 2\pi(x - vt + 1.5)cm$$

These waves are simultaneously passing through a string. The amplitude of the resulting wave is

(1) 2 cm (2) 4 cm

(3) 5.8 cm (4) 8 cm

PWA246

18. A longitudinal wave is represented by x = 10 sin $2\pi\left(nt-\frac{x}{\lambda}\right)$ cm. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to :

(1) 2π (2) 5π (3) π (4) $\frac{5\pi}{2}$

PWA247

19. The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is 1.2×10^{11} Nm^{-2}. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}$ m. The value of x is_____.

PWA248

20. The distance between two consecutive points with phase difference of 60° in a wave of frequency 500 Hz is 6.0 m. The velocity with which wave is traveling is_____ km/s

PWA249

21. The displacement equations of two interfering waves are given by

$y_1 = 10\sin\left(\omega t + \frac{\pi}{3}\right)$ cm,

$y_2 = 5\left[\sin(\omega t) + \sqrt{3}\cos\omega t\right]$cm respectively. The amplitude of the resultant wave is _____cm.

PWA250

22. A wire of length 'ℓ' having tension T and radius 'r' vibrates with fundamental frequency 'f'. Another wire of the same metal with length 2ℓ having tension 2T and radius 2r will vibrate with fundamental frequency :-

(1) f (2) 2f (3) $\frac{f}{2\sqrt{2}}$ (4) $\frac{f}{2}\sqrt{2}$

PWA251

23. A wave is represented by the equation $y = 10\sin2\pi(100t - 0.02x) + 10\sin2\pi(100t + 0.02x)$. The maximum amplitude and loop length are respectively: -

(1) 20 units and 30 units

(2) 20 units and 25 units

(3) 30 units and 20 units

(4) 25 units and 20 units

PWA252

24. In case of closed organ pipe which harmonic the pth overtone will be :-

(1) 2p + 1 (2) 2p – 1 (3) p + 1 (4) p – 1

PWA253

25. An organ pipe of length L is open at one end and closed at other end. The wavelengths of the three lowest resonating frequencies that can be produced by this pipe are :-

(1) 4L, 2L, L (2) 2L, L, L/2

(3) 2L, L, 2L/3 (4) 4L, 4L/3, 4L/5

PWA254

26. The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm. The length of the open organ pipe is ____ cm.

PWA255

27. A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n + 1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is…….kg/m.

PWA256

28. In an experiment to determine the velocity of sound in air at room temperature using a resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of ___cm.

PWA257

ANSWER KEY

1.	(2)	**2.**	(2)						
3.	(a) 6m, 0.25 Hz, 1.5 m/s, (b) 1.5π mm/s, $0.75\pi^2$ mm/s								
4.	(1)	**5.**	(4)	**6.**	(4)	**7.**	(3)	**8.**	(2, 4)
9.	(3)	**10.**	(3)	**11.**	(3)	**12.**	(1)	**13.**	(2)
14.	(2)	**15.**	(3)	**16.**	(1)	**17.**	(1)	**18.**	(2)
19.	15	**20.**	18	**21.**	20	**22.**	(3)	**23.**	(2)
24.	(1)	**25.**	(4)	**26.**	80	**27.**	3	**28.**	104

Units, Dimensions and Measurement

1. Match List I with List II

List		List II	
A.	Young's Modulus (Y)	I.	[M L^{-1} T^{-1}]
B.	Co-efficient of Viscosity (η)	II.	[M L^{2} T^{-1}]
C.	Planck's Constant (h)	III.	[M L^{-1} T^{-2}]
D.	Work Function (ϕ)	IV.	[M L^{2} T^{-2}]

Choose the correct answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-III, C-IV, D-II
(4) A-I, B-II, C-III, D-IV

PPM175

2. Match List I with List II :

List-I (Physical Quantity)		List-II (Dimensional Formula)	
A	Pressure gradient	I	$[M^0L^2T^{-2}]$
B	Energy density	II	$[M^1L^{-1}T^{-2}]$
C	Electric Field	III	$[M^1L^{-2}T^{-2}]$
D	Latent heat	IV	$[M^1L^1T^{-3}A^{-1}]$

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

PPM176

3. Which of the following is not correct representation of a unit?
(1) Newton (2) Sec
(3) a.m.u. (4) All of these

PPM177

4. Which one of the following is unitless quantity?
(1) Velocity gradient
(2) Pressure gradient
(3) Displacement gradient
(4) Force gradient

PPM178

5. An expression for a dimensionless quantity P is given by $P=\frac{\alpha}{\beta}\log_e\left(\frac{kT}{\beta x}\right)$; where α and β are constants, x is distance ; k is Boltzmann constant and T is the temperature. Then the dimensions of α will be :
(1) [M^{0}L^{-1}T^0] (2) [ML^{0}T^{-2}]
(3) [MLT^{-2}] (4) [ML^{2}T^{-2}]

PPM179

6. The potential energy of a particle varies with distance x from a fixed origin as $U=\frac{A\sqrt{x}}{x^2+B}$, where A and B are dimensional constants then dimensional formula for AB is
(1) [ML$^{7/2}$ T^{-2}] (2) [ML$^{11/2}$ T^{-2}]
(3) [M^{2}L$^{9/2}$ T^{-2}] (4) [ML$^{13/2}$ T^{-3}]

PPM180

7. In Vander Waals equation

$$\left[P+\frac{a}{V^2}\right][V-b]=RT;$$

P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to :
(1) $\frac{P}{V}$ (2) $\frac{V}{P}$
(3) PV (4) PV^3

PPM181

8. Which of the following relation cannot be deduced using dimensional analysis? [the symbols have their usual meanings]
(1) y = A sin(ωt + kx) (2) v = u + at
(3) k = $\frac{1}{2}$ mv^2 (4) All of these

PPM182

9. If momentum (P), area (A) and time (T) are taken to be fundamental quantities, then energy has the dimensional formula :
(1) [P^{1}A^{-1}T^1] (2) [P^{2}A^{1}T^1]
(3) [P^{1}A$^{-1/2}$T^1] (4) [P^{1}A$^{1/2}$T^{-1}]

PPM183

10. If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

(1) $[P\ A^{-1}\ T^{0}]$ (2) $[P\ A\ T^{-1}]$

(3) $[P\ A^{-1}\ T]$ (4) $[P\ A^{-1}\ T^{-1}]$

PPM184

11. If a = 8 ± 0.08 and b = 6 ± 0.06, Let x = a + b, y = a – b, z = a × b. The correct order of % error in x, y and z

(1) x = y < z (2) x = y > z

(3) x < z < y (4) x > z < y

PPM185

12. Two resistance are measured in ohm and is given as: -

$R_1 = 3\Omega \pm 1\%$

$R_2 = 6\Omega \pm 2\%$

When they are connected in parallel, the percentage error in equivalent resistance is

(1) 3% (2) 4.5%

(3) 0.67% (4) 1.33%

PPM186

13. The number of significant digits in 0.001001 is

(1) 6 (2) 4 (3) 7 (4) 2

PPM187

14. The sides of a rectangle are (10.5 ± 0.2) cm and (5.2 ± 0.1) cm. Then its perimeter with error limits is :-

(1) (31.4 ± 0.6)cm (2) (31.4 ± 0.3)cm

(3) (51.6 ± 0.6)cm (4) (51.6 ± 0.3)cm

PPM188

15. For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12%, respectively, then the percentage error for 'z' will be %.

PPM189

16. An aluminium wire is stretched to make its length, 04% larger. Then percentage change in resistance is:

(1) 0.4 % (2) 0.2 %

(3) 0.8 % (4) 0.6 %

PPM190

17. A sliver wire has mass (0.6 ± 0.006) g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be :

(1) 4% (2) 3% (3) 6% (4) 7%

PPM191

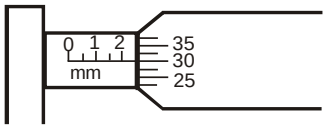
18. The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be:

(1) 2% (2) 4%

(3) 6% (4) 8%

PPM192

19. What is the reading of micrometer screw gauge shown in figure



(1) 2.30 mm (2) 2.29 mm

(3) 2.36 mm (4) 2.41 mm

PPM193

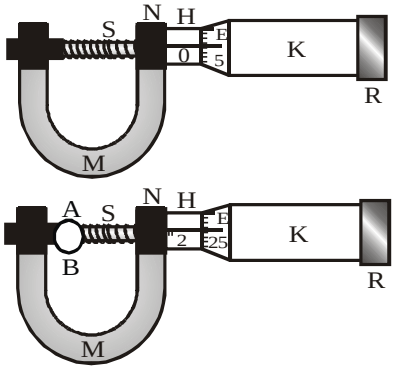
20. In a vernier calliper, N divisions of vernier scale coincide with (N – 1) divisions of main scale (in which 1 division represents 1mm). The least count of the instrument in cm. should be

(1) N (2) $\frac{1}{N}$

(3) $\frac{1}{10N}$ (4) $\frac{1}{N-1}$

PPM194

21. The circular scale of a screw gauge has 50 divisions and pitch of 0.5 mm. Main scale reading is 2. Find the diameter of sphere.



(1) 1.2 mm (2) 1.25 mm

(3) 2.20 mm (4) 2.25 mm

PPM195

22. The vernier of a circular scale is divided into 30 divisions which coincide against 29 divisions of main scale. Each main scale division is 0.5°. The least count of the instrument is –

(1) 10' (2) 0.1' (3) 1' (4) 30'

PPM196

23. The pitch of a screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a wire, the linear scale reads 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. Find the curved surface area (in cm^2) of the wire in appropriate number of significant figures.

PPM197

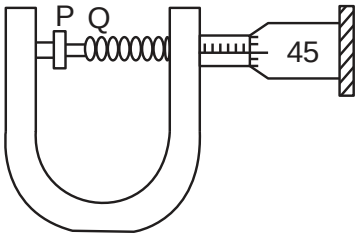
24. The vernier constant of Vernier callipers is 0.1 mm and it has zero error of (–0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be____ × 10^{-2} cm.

PPM198

25. The one division of main scale of vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6th Vernier division coincides with a main scale division. The diameter of the bob will be____ 10^{-2} cm.

PPM199

26. In an experiment to find out the diameter of wire using screw gauge, the following observation were noted :



(a) Screw moves 0.5 mm on main scale in one complete rotation

(b) Total divisions on circular scale = 50

(c) Main scale reading is 2.5 mm

(d) 45th division of circular scale is in the pitch line

(e) Instrument has 0.03 mm negative error

Then the diameter of wire is :

(1) 2.92 mm (2) 2.54 mm

(3) 2.98 mm (4) 3.45 mm

PPM200

ANSWER KEY

1.	(2)	**2.**	(3)	**3.**	(4)	**4.**	(3)	**5.**	(3)
6.	(2)	**7.**	(3)	**8.**	(4)	**9.**	(4)	**10.**	(1)
11.	(3)	**12.**	(4)	**13.**	(2)	**14.**	(1)	**15.**	(18)
16.	(3)	**17.**	(1)	**18.**	(4)	**19.**	(1)	**20.**	(3)
21.	(1)	**22.**	(3)	**23.**	2.6 cm^2	**24.**	180	**25.**	412
26.	(3)								